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**Identifying Galaxy Cluster Members with Machine Learning out to
Large Cluster-centric Distances**

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Abstract

Galaxy clusters are the largest gravitationally bound structures in the Universe and serve as powerful laboratories for studying galaxy evolution in dense environments. However, accurately identifying cluster members is a challenging task due to projection effects and the presence of interlopers, particularly in the cluster outskirts beyond the virialized core. This work addresses the membership classification problem out to large cluster-centric distances ($5 r_{200}$) using a novel Machine Learning approach within the context of the CHANCES survey.

We utilize CHANCES mock catalogs constructed on the Uchuu simulation to train a Random Forest classifier, employing a sample of 42 massive clusters ($M_{200} > 10^{14} M_{\odot}$) at low redshift ($z < 0.07$). To address the inherent class imbalance, we evaluate six model variants by combining three resampling strategies (Original Data, SMOTE and RandomUnderSampler) with and without hyperparameter optimization, training the algorithms on Projected Phase Space information (v_{pec} , r_{proj}) combined with local density (Σ_{10}) and intrinsic galaxy and halo information (m_r , M_{200}).

Our best-performing model achieves an overall Completeness of 93% and a Purity of 66%, maintaining an optimal balance. We observe a radial dependence in performance, where Purity naturally decreases towards the outskirts due to severe background contamination. The analysis reveals that the algorithm relies heavily on dynamical features, behaving similarly to the traditional caustic technique and yielding competitive results with state-of-the-art approaches. Our findings demonstrate that Machine Learning is a robust tool for this task, effectively recovering members at large radii, where the limitations are primarily physical—driven by projection effects—rather than algorithmic, establishing a reliable framework for future analyses of the CHANCES spectroscopic data.

Resumen

Los cúmulos de galaxias son las estructuras gravitacionalmente ligadas más grandes del Universo y actúan como poderosos laboratorios para estudiar la evolución de las galaxias en entornos densos. Sin embargo, identificar con precisión los miembros del cúmulo es una tarea desafiante debido a los efectos de proyección y la presencia de intrusos, particularmente en las afueras del cúmulo, más allá del núcleo virializado. Este trabajo aborda el problema de la clasificación de miembros hasta grandes distancias cúmulo-céntricas ($5 r_{200}$) utilizando un enfoque de Machine Learning en el contexto del sondeo CHANCES.

Utilizamos catálogos simulados de CHANCES contruidos a partir de la simulación Uchuu para entrenar un clasificador Random Forest, empleando una muestra de 42 cúmulos masivos ($M_{200} > 10^{14} M_{\odot}$) a bajo redshift ($z < 0.07$). Para abordar el desequilibrio de clases inherente, evaluamos seis variantes del modelo combinando tres estrategias de remuestreo (Datos originales, SMOTE, y RandomUnderSampler) con y sin optimización de hiperparámetros. Los algoritmos se entrenan utilizando información del Espacio de Fase Proyectado (v_{pec}, r_{proj}) combinada con la densidad local (Σ_{10}) e información intrínseca de las galaxias y halos (m_r, M_{200}).

Nuestro mejor modelo logra una Completitud general del 93% y una Pureza del 66%, manteniendo un equilibrio óptimo. Observamos una dependencia radial en el rendimiento, donde la Pureza disminuye naturalmente hacia las afueras debido a la severa contaminación de fondo. El análisis revela que el algoritmo depende fuertemente de características dinámicas, comportándose de manera similar a la técnica tradicional de cáusticas y arrojando resultados competitivos con el estado del arte. Nuestros hallazgos demuestran que el Machine Learning es una herramienta robusta para esta tarea, recuperando eficazmente miembros a grandes distancias, donde las limitaciones son principalmente físicas —impulsadas por efectos de proyección— más que algorítmicas, estableciendo un marco confiable para futuros análisis de los datos espectroscópicos de CHANCES.

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Chapter 1

Introduction

In modern cosmology, the Λ CDM (Lambda cold dark matter) model has established the paradigm of hierarchical structure formation. In this framework, the Universe evolves from primordial density fluctuations generated in the early Universe into progressively larger structures through gravitational instability (Zel’dovich 1970; White and Rees 1978; Peebles 1980). Dark matter halos collapse first, subsequently merging and accreting mass over cosmic timescales, generating the gravitational potential wells where baryons eventually cool off and condense to form galaxies. As these structures grow, they assemble into the massive nodes of the cosmic web that we observe today (Kravtsov and Borgani 2012).

In this hierarchical context, galaxy clusters represent the largest gravitationally bound structures in the Universe, composed of hundreds up to thousands of galaxies embedded in a dominant dark matter halo ($\sim 80 - 85\%$ of their mass) and permeated by a hot, diffuse Intracuster Medium that strongly emits in X-rays (Schneider 2015). As the massive nodes of the cosmic web, galaxy clusters serve as powerful natural laboratories for cosmological model testing (Allen et al. 2011), understanding large-scale structure formation (Kravtsov and Borgani 2012) and investigating the environmental mechanisms that drive galaxy evolution (Boselli and Gavazzi 2006; Wetzel et al. 2013; C. P. Haines et al. 2015). However, any quantitative analysis of cluster properties, ranging from mass profiles and velocity dispersion to luminosity functions and environmental transformation of galaxies, requires a reliable identification of the galaxies that are truly bound to the system. Thus, the challenge of determining cluster membership is one of the fundamental steps in any cluster-based study.

1.1 Galaxy Cluster Membership Classification

Identifying which galaxies genuinely belong to a cluster is a fundamental yet challenging task in observational extragalactic astrophysics. Accurate membership classification is critical, as any ambiguity directly impacts the derivation of key cluster properties. In real observations, cluster membership cannot be measured directly in three-dimensional space; instead, we must rely on projected positions on the sky (RA, DEC) and line-of-sight velocities derived from redshifts (z).

This projection introduces a fundamental degeneracy: the observed redshift combines the cosmological expansion (Hubble flow) with the galaxy’s own peculiar velocity (v_{pec}) along the line of sight. Consequently, galaxies that are not physically associated with the cluster may appear pro-

jected along the same direction (Rozo et al. 2015), contaminating the sample. These foreground and background contaminants, commonly referred to as *interlopers*, can significantly bias measurements (Wojtak et al. 2007; Mamon et al. 2010; Saro et al. 2013). Even a modest level of contamination can artificially inflate the velocity dispersion (σ_v), leading to overestimated dynamical masses, or cover dynamical features relevant to reconstruct the cluster’s assembly history.

1.1.1 Traditional Approaches and Limitations

Traditionally, membership selection has relied on methods that exploit specific physical or statistical properties of cluster galaxies, yet each comes with distinct limitations, particularly at large radii:

- **Photometric Methods (The Red Sequence):** Historically, one of the most efficient methods for detecting clusters and selecting members has been the Red Sequence technique (Gladders and Yee 2000). This relies on the observation that the dense cores of galaxy clusters are dominated by old, passive elliptical galaxies, following a morphology-density relation (Dressler 1980). However, this method is biased against the star-forming, blue galaxy population. This bias becomes critical in higher redshift orders, where the fraction of core blue, star-forming galaxies increases due to the Butcher-Oemler effect (Butcher and Oemler 1984). Furthermore, relying solely on color cuts could potentially exclude the infall regions, systematically missing the very population undergoing environmental transformation (quenching) as they enter the cluster.
- **Dynamical Methods (Projected Phase Space):** Spectroscopic surveys allow for selection based on redshift space, typically using iterative σ -clipping (Yahil and Vidal 1977) or the caustic technique (Diaferio and Geller 1997; Diaferio 1999), which operate in the Projected Phase Space (PPS; line-of-sight velocity versus cluster-centric radius). While robust in the virialized core ($r < r_{200}$, where r_{200} is the virial radius, enclosing a mean density of 200 times the critical density of the Universe ρ_c), in the non-virialized outskirts, infalling groups and filaments introduce complex kinematic structures that deviate from simple dynamical equilibrium models, making it difficult to distinguish members from the field population in the higher cluster-centric distance regime based on velocity cuts alone, especially in under-sampled clusters.

As illustrated in Figure 1.1, the PPS contains distinct dynamical populations. Virialized galaxies occupy the cluster core and have settled into dynamical equilibrium. Infalling galaxies correspond to those falling into the potential well for the first time, exhibiting coherent infall trajectories. A particularly challenging population are the back-splash galaxies: objects that have already passed through the cluster pericenter but have not necessarily reached the apocenter of their orbits, effectively "splashing back" out to radii as large as $2 - 3 r_{200}$ (C. P. Haines et al. 2015; Haggard et al. 2020). While distinct in the six-dimensional phase space, in the PPS, back-splash galaxies often share the same locus as the infalling population (Rhee et al. 2017).

Most galaxy populations in clusters studies conducted to date have focused primarily on the virialized regions or, at most, extended out to $\sim 3r_{200}$, as seen in surveys like WINGS (Fasano et al. 2006), OmegaWINGS (Gullieuszik et al. 2015) and LoCUSS (e.g., Richard et al. 2010; C. P. Haines et al. 2015). However, as we know from Λ CDM, galaxies are being "pre-processed" in intermediate-density environments, such as groups and filaments, prior to cluster infall (Zabludoff et al. 1996; C. P. Haines et al. 2015; P. A. Lopes et al. 2024). The increasing interest in this field has led to

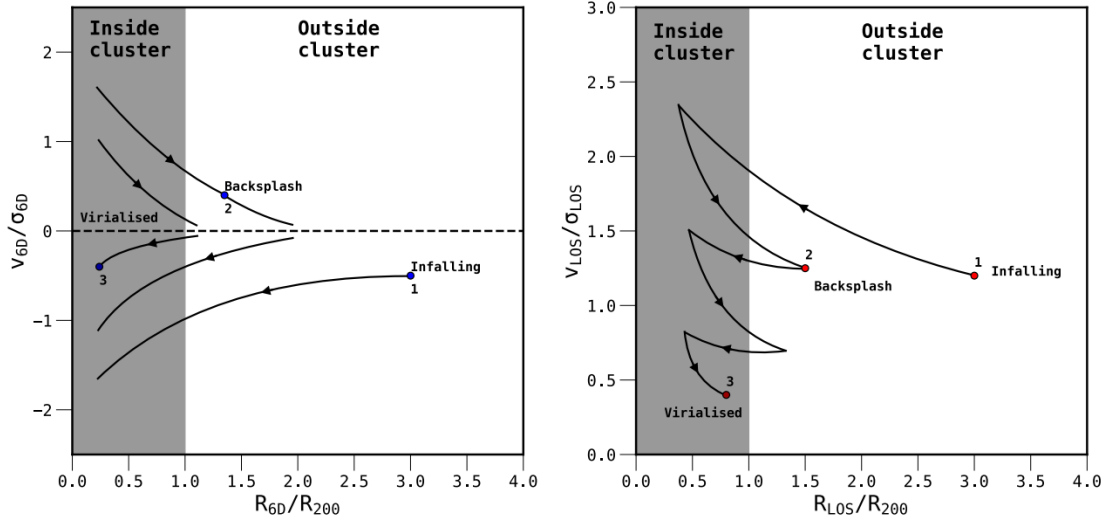


Figure 1.1: A schematic view of the evolution of a halo or subhalo in the Phase Space Diagram from Arthur et al. (2019). *Left Panel:* The "true" six-dimensional $(\vec{r}, \vec{v})$ phase space. *Right Panel:* The projected phase space $(v_{l.o.s} \text{ vs } r_{l.o.s})$. The diagram shows the dynamical complexity of galaxy clusters, distinguishing between three main populations: virialized members, located deep in the gray-colored potential well; infalling galaxies, that are entering the cluster for the first time; and back-splash galaxies, that have passed through the core ($r < r_{200}$) at least once but are currently located in the outskirts ($r > r_{200}$) due to their orbital trajectory.

the suggestion that a significant fraction of galaxy transformation takes place not just inside the clusters themselves, but in these surrounding large-scale structures out to at least $5 r_{200}$, where the signatures of pre-processing are expected to be recognizable (Bahé et al. 2013; P. A. Lopes et al. 2024; C. Haines et al. 2023) and where the turn-around radius is expected to be found (Bertschinger 1985; Rines and Diaferio 2006), that is, the radius at which galaxies detach from the Hubble flow and become gravitationally bound to the cluster.

1.1.2 The Challenge of the Outskirts: The CHANCES Survey

The difficulty of membership classification worsens significantly at large cluster-centric distances (Moretti et al. 2017). Beyond the virialized core, the density contrast between genuine cluster members and field galaxies becomes increasingly low.

Addressing this observational gap is the primary motivation for the upcoming **CHANCES** (CHileAN Cluster galaxy Evolution Survey) project¹, which is one of the 9 4MOST Community surveys (de Jong et al. 2019) that will be carried over its first 5 years of operation, starting in 2026. CHANCES aims to study the evolution of galaxies not just in the cluster cores, but extending out to $5 r_{200}$ (C. Haines et al. 2023; Sifón et al. 2025; Baier-Soto et al. 2025; Méndez-Hernández et al. 2025). By obtaining spectroscopy for $\sim 500,000$ galaxies in and around ~ 150 massive clusters, CHANCES will map the pre-processing of galaxies within groups and filaments before they reach the main cluster potential. This unprecedented wide-field coverage will help to trace the relation between the evolution of galaxies and hierarchical structure formation as it occurs.

¹chances.uda.cl

CHANCES is structurally divided into three sub-surveys to cover distinct epochs of cluster evolution: the Low- z , the Evolution and the CGM sub-surveys. Particularly, the Low- z CHANCES sub-survey will target galaxies in and around 50 galaxy clusters at $z < 0.07$ and with a mass range of around $10^{14} M_{\odot} \lesssim M_{200} \lesssim 10^{15} M_{\odot}$, in addition to the Shapley ($z = 0.048$) and Horologium-Reticulum ($z = 0.060$) supercluster regions (Sifón et al. 2025). Utilizing the high multiplexing capabilities of the 4MOST instrument, the survey will reach a limiting magnitude of $r_{AB} < 20.4$. This depth allows for the detection of dwarf galaxies with stellar masses as low as $10^{8.5} M_{\odot}$, providing an unprecedented view of the faint galaxy population across a wide field of view.

CHANCES will offer a new window to galaxy evolution due to the wide and deep approach. However, extending the usual cluster analysis to $5 r_{200}$ amplifies the contamination challenge. At these radii, the distinct "trumpet" shape (Kaiser 1987; Regos and Geller 1989) of the caustics in the phase space merges into the Hubble flow. Traditional methods, often based on iterative cuts or static velocity limits, struggle to exclude interlopers while retaining infalling members. Therefore, to fully exploit the wide-field capabilities of CHANCES and accurately map the accretion of galaxies, a more flexible classification technique is required; one capable of learning the optimal decision boundaries within the PPS and adapting to the complex dynamics of cluster outskirts in order to distinguish members from interlopers where dynamical cuts fall short.

1.2 Machine Learning in Astrophysics

As we enter the era of large-scale surveys, such as LSST (Vera C. Rubin Observatory), Euclid, and 4MOST, the exponential growth of astronomical data has triggered the need for automated, scalable pipelines that allow the processing of the generated volume. The resulting "data deluge" renders manual analysis or traditional algorithms insufficient for processing millions of sources efficiently and accurately. In this context, **Machine Learning (ML)** has emerged as an indispensable tool for automation and pattern recognition within high-dimensional, heterogeneous datasets, enabling the extraction of physical insights from complex patterns that may otherwise be hardly accessible by standard analytical techniques.

Machine Learning algorithms have proven their utility not only in the industry but across various fields of astrophysics (Baron 2019; Kembhavi and Pattnaik 2022), demonstrating their automation capabilities. Early efforts, such as the work conducted by De La Calleja and Fuentes (2004) explored automated morphological classification employing Neural Networks and Locally Weighted Regression ensembles. Although limited by the small dataset employed at the time (301 galaxies), they established the feasibility of automated classification, achieving high accuracy rates. In the modern era, these techniques have become more sophisticated; recently, Kolesnikov et al. (2023) proposed a hybrid method combining unsupervised clustering (SOMBRERO) with a supervised Convolutional Neural Network (CNN). By using unsupervised Machine Learning to refine the training labels, they achieved an overall accuracy of $\sim 95\%$ in the best-case scenario, demonstrating the power of hybrid architectures for handling larger, complex data. Alternatively, Aguilar-Argüello et al. (2025) employed XGBoost for morphological classification using a set of 14 structural and star-formation parameters. Their model achieved an accuracy of 88% for binary Early Type vs Late Type classification, though this performance dropped to 65% when subdividing galaxies into five morphological categories, highlighting the challenge of detailed multi-class separation using photometric features.

alone.

Beyond morphology, Lochner et al. (2016) demonstrated that feature-based ML classifiers such as Boosted Decision Trees, could distinguish between different types of supernovae using photometric light curves alone, a crucial step for surveys like LSST, where spectroscopic follow-up for all targets is nearly impossible. Similarly, in cosmology, Pasquet et al. (2019) showed that CNNs applied directly to galaxy images could estimate photometric redshifts with a precision rivaling or exceeding standard template-fitting methods. Furthermore, in exoplanet astrophysics, Dattilo et al. (2019) utilized Deep Learning to identify subtle transit signals in Kepler data that had been missed by previous automated pipelines, significantly expanding the candidate catalog.

Moving on to the galaxy cluster membership assignment, supervised and non-supervised learning algorithms also offer a promising alternative to traditional membership selection. For instance, P. A. A. Lopes and Ribeiro (2020) addressed the membership challenge from a purely photometric perspective, aiming to bypass the limitations of expensive and scarce spectroscopic data. They implemented a Support Vector Machine (SVM) algorithm trained on SDSS data to classify members based exclusively on galaxy colors and magnitudes, establishing a "Reliable Photometric Membership" method that performs robustly in the local Universe. Focusing on the kinematic domain, Farid et al. (2023) introduced the C²-GAME framework, which utilizes a suite of ML models trained on mock catalogs from MDPL2 N-body simulation. By leveraging PPS information up to $5 r_{200m}$, they demonstrated the ability to probabilistically differentiate between orbiting members, infalling galaxies and background interlopers. However, this framework is primarily designed to derive unbiased statistical estimates of global cluster properties, such as velocity dispersion and density profiles. While effective for these aims, this approach prioritizes the global statistical accuracy over the precision of individual classification. The evolutionary studies such as CHANCES, in contrast, require a highly accurate membership catalog to reliably trace the environmental dependence of galaxy properties. In this regard, there remains a need for a classification strategy that is explicitly optimized to isolate the true cluster population out to large cluster-centric distances, enabling the robust analysis of the faint, pre-processed galaxy population.

1.3 This thesis

With the intent to bridge this gap, this work focuses on implementing a supervised Machine Learning approach based on a Random Forest classifier to identify galaxy cluster members within the CHANCES framework. The algorithm selection contemplated its robustness against overfitting and its ability to model non-linear decision boundaries in multi-dimensional feature spaces, essential characteristics for disentangling the overlapping distributions of interlopers and members. By training on realistic mock catalogs, the model empirically learns to optimize this complex separation. Our primary goal is to optimize the trade-off between Purity and Completeness, particularly extending the analysis out to large cluster-centric distances ($5 r_{200}$). This regime, which encompasses the infall zone and the connection with the cosmic web, is where contamination from field galaxies is most severe and where previous Machine Learning studies have been limited. By developing and calibrating a robust classification tool, tailored to the CHANCES Low- z survey specifics, this thesis will serve as the foundation to accurately map the accretion zones of clusters, thereby enabling future studies on the environmental quenching, pre-processing and assembly histories of galaxies across the full

dynamic range encompassed by a cluster’s sphere of influence.

The structure of this thesis is as follows: Chapter 2 describes the Data used in this work, detailing the CHANCES survey specifications and the mock simulations used for training. Chapter 3 presents the Methodology, covering the labeling phase, as well as the feature engineering process, the training strategies and the optimization of the Random Forest model. Chapter 4 reports the classification performance and validation results. Chapter 5 provides a Discussion of these results, analyzing the physical implications as well as the physical and algorithmic limitations. Finally, Chapter 6 summarizes the Conclusions, followed by Chapter 7, which outlines avenues for future work.

Chapter 2

Data: The CHANCES Mock Catalogs

The development of a supervised Machine Learning algorithm, such as Random Forest, requires a labeled dataset that accurately portrays the "ground truth". In observational astrophysics, we lack access to the full six-dimensional phase space of galaxies. We are limited to projected coordinates (RA, DEC) and the line-of-sight velocity, which inevitably introduces contamination and therefore hinders the distinction between gravitationally bound and field galaxies.

To overcome this fundamental limitation, and since traditional membership classifications are themselves estimates and may contain biases, cosmological simulations provide an alternate solution to this problem, offering:

- **Ground Truth:** In simulations, the full 3D potential is well known. We can define a "true member" physically as a galaxy that is gravitationally bound to the cluster halo, regardless of its projected position.
- **Controlled Environment:** Simulations allow us to test the model in a controlled setting, ensuring that the algorithm learns physical patterns rather than observational artifacts.

To leverage these advantages, this work utilizes a specific suite of realistic mock catalogs designed to support the CHANCES Low- z survey.

The mocks were generated to reproduce the survey's spatial, photometric, mass selection, and redshift properties, including observational uncertainties and selection effects. They serve as the testing facility for our algorithm, providing the necessary ground truth to validate our model's predictions.

2.1 Simulation Suite and Galaxy Population

The CHANCES mocks are derived from the Uchuu simulation (Ishiyama et al. 2021), a high-resolution, large-scale cosmological N-body simulation that provides the dark-matter-halo backbone. Uchuu offers a unique combination of large volume and high mass resolution, essential for resolving the sub-halos that host the galaxy populations relevant to CHANCES down to dwarf-galaxy scales.

Since this N-body simulation only traces dark matter, the halos were populated with galaxies using the UNIVERSE MACHINE empirical model (Behroozi et al. 2019). This algorithm connects

galaxy formation physics to the growth of dark matter halos, assigning properties such as Star Formation Rate (SFR) and stellar mass (M_*) to the simulated halos in a way that reproduces observed scaling relations (e.g., the Stellar Mass Function and SFR- M_* relation) across cosmic time. The resulting catalog is complete down to $M_* \sim 10^8 M_\odot$, close to the spectroscopic limit of the CHANCES Low- z survey ($M_* > 10^{8.5} M_\odot$) (Sifón et al. 2025).

2.2 Target Selection: Finding Cluster Analogs

For each target in the CHANCES Low- z catalog, a systematic search was conducted within the full-sky mocks to identify a simulated cluster that matched its key physical attributes. Specifically, the selection prioritized candidates with a concordant redshift within 0.01 and a matching virial mass M_{200} and virial radius r_{200} within 0.01 dex, ensuring that the mock cluster replicates the gravitational potential depth and velocity dispersion of the observed systems. This process ensures that the training set covers the same parameter space as the real survey, minimizing biases when the model is applied to observational data.

2.3 Light-cone Construction and Photometry

To transform the simulation snapshots into a format comparable to that of observational surveys, light-cones were constructed using the methodology described by (Comparat et al. 2020). This process involves extracting spherical shells from simulation snapshots and concatenating them along the line of sight according to the redshift of the structures. By arranging these according to their comoving distance, the method replicates the geometry of the observable Universe.

To ensure the training set is representative of the real CHANCES targets (defined in Sifón et al. 2025), the generated light-cones were centered on the simulated cluster analogs identified.

Additionally, synthetic photometry was assigned to the simulated galaxies to mimic the specific instrument responses expected for the survey targets. Following the procedure detailed in Comparat, (in prep.), observed Spectral Energy Distributions from reference surveys—such as GAMA (Driver et al. 2012), KiDS (Hildebrandt et al. 2021) and COSMOS (Ilbert et al. 2013)—were matched to the simulated galaxies based on their physical properties. This allows the generation of realistic apparent magnitudes (e.g., g, r, i, z) and colors, naturally reproducing the blue cloud and the characteristic red sequence, which were explicitly modeled using calibrations from the redMaPPer cluster finder (Rykoff et al. 2014). Furthermore, spectroscopic redshifts (‘Redshift_S’) were assigned by adding the galaxies’ peculiar velocities projected along the line of sight to the galaxies’ real redshifts (‘Redshift_R’).

2.4 Final Selection and Quality Cuts

The initial suite consisted of 53 mock clusters. It is worth noting that this number exceeds the official CHANCES Low- z sample size (see Sifón et al. 2025) because the mocks were generated using a preliminary target list that included three additional clusters, which were later removed from the survey definition.

To ensure a robust training set for the Machine Learning algorithm, a selection process was applied to the initial 53 mock pool. This filtering reduced the sample to a final set of **42 mocks**,

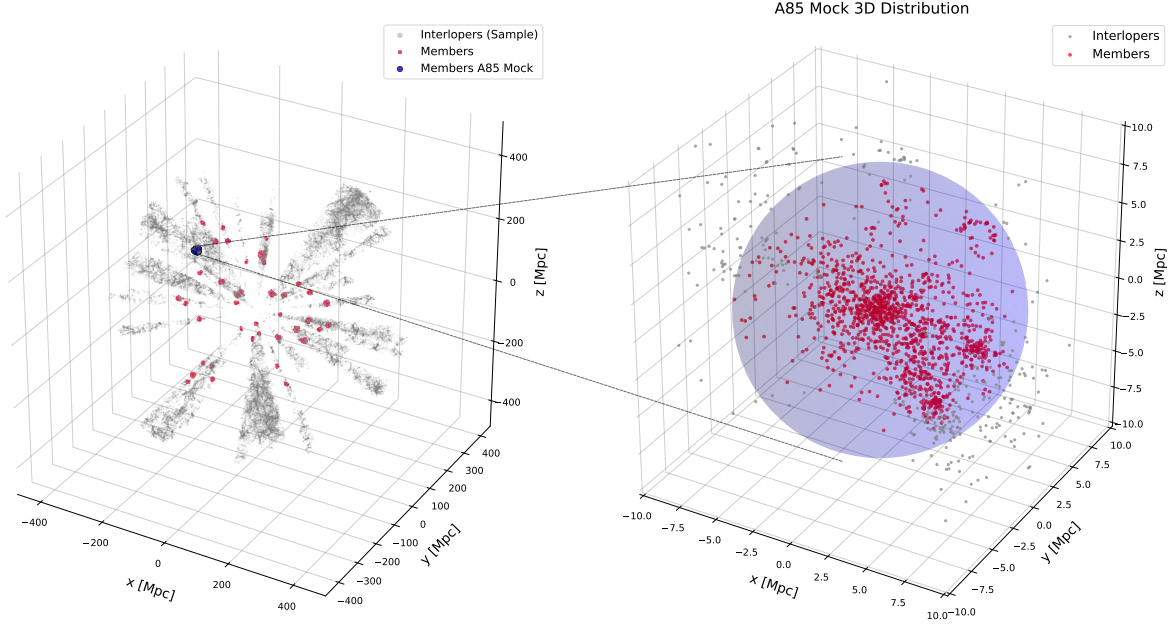


Figure 2.1: Spatial distribution of the 42 mock clusters used in this work. *Left panel*: the full distribution of the mock light-cones in projected sky coordinates, with red and blue (Mock A85) dots indicating the location of galaxy clusters. *Right panel*: a zoom-in on the Abell 85 cluster (Mock 0), where red points represent gravitationally bound members within $5 r_{200}$ and gray points represent interlopers. This 3D information allows for the precise labeling required for supervised training.

based on the following criteria:

- **Removal of redundant clusters:** Due to the possibility of two similar clusters with alike z , M_{200} being mapped to the same simulated halo in the light-cone, including the same mock twice would artificially inflate the performance of the classifier. As a result, we identified and removed these duplicate realizations: **Abell 1520** (identical to Abell 970), **Abell 3571** (identical to Abell 119), and both **Abell 3581** and **Hydra (Abell 1060)**, which were duplicates of Abell 3565.
- **Member Threshold ($N > 150$):** To ensure sufficient statistical significance for the training process, only clusters with over 150 galaxies within $5 r_{200}$ were retained. This selection is illustrated in Figure 2.2, which displays the ordered member counts for the initial mock suite. Mocks with fewer members (light blue) were not deemed sufficient to adequately populate the PPS, limiting the algorithm’s decision-making capabilities. Following this criterion, sparse systems such as **Abell 194**, **Abell 3389**, **Abell 3809**, **Abell S560** and **Fornax** were excluded.
- **Dynamical State Anomalies:** A visual inspection of the PPS and the 3D spatial distribution was conducted for each mock (see Appendix C). Systems exhibiting highly anomalous phase-space structures, which could introduce excessive noise into the training process, were excluded. The latter resulted in the removal of **Abell 500** and **Abell 3266**, both identified as cluster mergers. Although the goal is to train a good generalizer, it is also important to train the model on a representative sample of clusters with reasonably defined caustics.

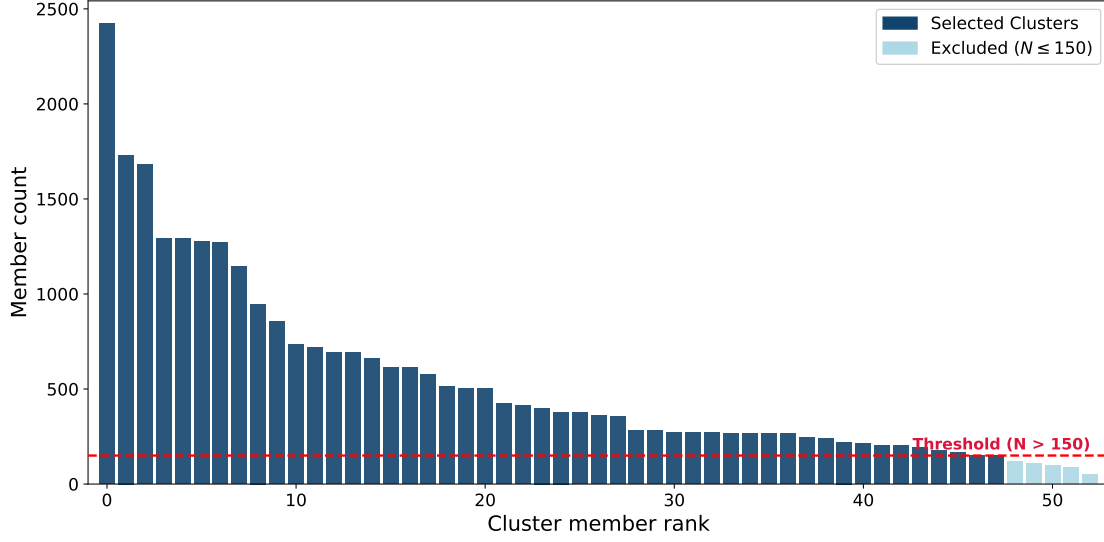


Figure 2.2: Descending sorted distribution of member counts within $5 r_{200}$ for the initial suite of 53 mock clusters. The dark blue bars represent the 48 clusters that passed the $N > 150$ criterion (dashed red line), while the light blue bars indicate the systems excluded by this threshold. The cut ensures that all clusters are statistically significant.

After applying these quality cuts and the redshift selection $z < 0.1$, the final dataset comprises a total of **282,523 galaxies**. This extensive catalog serves as input for training and validating our Machine Learning models.

Figure 2.1 provides an overview of the spatial distribution of the 42 selected mock light-cones, including a detailed view of the **Abell 85** mock to illustrate the distinction between members and interlopers in 3D space. The characteristics of the final 42 selected mocks are detailed in Table B.1. Additionally, the Projected Phase Space diagrams and 3D spatial distributions for each selected mock are presented in Appendix C.

Chapter 3

Methodology

We employ a supervised Machine Learning approach to identify galaxy cluster members extending out to $5 r_{200}$; a critical region for understanding galaxy evolution, but heavily contaminated by interlopers, as described in Chapter 1. Unlike methods that rely on static cuts, supervised ML algorithms can learn the complex mapping between observables and membership directly from the data. However, for a supervised model to learn effectively, we require a labeled dataset where the ground truth is known with certainty. As detailed in Chapter 2, we make use of realistic mock catalogs from cosmological simulations, where the 3D phase space information allows us to tag galaxies as "Members" or "Interlopers" with certainty.

3.1 Defining Membership in Simulations

A major advantage when using simulated mock catalogs is the access to the full six-dimensional phase space ($\vec{r}$, $\vec{v}$) of every galaxy, allowing for a precise definition of cluster membership. Profiting from this reality, we adopt a geometric definition of membership based on the three-dimensional cluster-centric distance, r_{3D} :

$$r_{3D} = \sqrt{(X - X_{cl})^2 + (Y - Y_{cl})^2 + (Z - Z_{cl})^2} \quad (3.1)$$

Where X , Y , Z are the galaxies' spatial coordinates and X_{cl} , Y_{cl} , Z_{cl} correspond to the cluster's central spatial coordinates. Based on this metric, we define our target labels as follows:

1. **Members (Class 1):** Assuming the cluster to be a three-dimensional sphere of radius $5 r_{200}$ (see right panel on Figure 2.1), any galaxy located within this volume is labeled as a member:

$$r_{3D} \leq 5 r_{200} \quad (3.2)$$

2. **Interlopers (Class 0):** Every other galaxy within the light-cone field of view that is physically located outside the three-dimensional sphere of the cluster:

$$r_{3D} > 5 r_{200} \quad (3.3)$$

These objects constitute foreground and background galaxies dominated by the Hubble flow, and in some cases, may appear spatially coincident with the cluster due to projection effects.

The instated binary classification scheme establishes the ground truth needed to train a supervised Machine Learning algorithm, allowing us to recover not only the virialized core of clusters, but also the infalling populations and filamentary structures feeding it in a single category. The resulting spatial distribution is illustrated in the right panel of Figure 2.1), where the $5 r_{200}$ sphere cleanly separates the "Members" category from the outside "Interlopers".

The geometric cut yields a significantly imbalanced dataset, a critical factor that needs to be taken into account in ML model training. The final labeled catalog contains:

- **True Members:** 23,372 galaxies ($\sim 8.3\%$)
- **Interlopers:** 259,151 galaxies ($\sim 91.7\%$)

The overwhelming prevalence of interlopers (with a $\sim 11 : 1$ ratio) constitutes a major challenge for classification algorithms, which tend to be biased towards the majority class. The specific strategies followed to sort this issue will be discussed in Section 3.6.3.

3.2 The Phase Space dilemma

Once membership is defined in 3D space, we can analyze how these populations project onto the observable 2D plane. Figure 3.1 presents the stacked Projected Phase Space for all 42 clusters, normalized by their respective virial radius r_{200} and velocity dispersion σ_{200} . The left panel displays the distribution of all galaxies in the catalog, where the members (red) form the expected trumpet shape (Kaiser 1987; Regos and Geller 1989). This characteristic distribution is a direct dynamical consequence of the cluster's gravitational potential: deep within the potential well ($r < r_{200}$), galaxies can sustain high orbital velocities, resulting in a large velocity dispersion (the "mouth" of the trumpet). As we move to larger cluster-centric distances, the gravitational influence weakens and thus the escape velocity decreases, narrowing the allowed range of velocities for bound members, ultimately forming the "body" of the trumpet.

While the positive class follows theoretical expectations, the observational challenge becomes evident when we isolate the interloper class (see right panel on Figure 3.1). Due to projection effects, the unbound galaxies, dominated by the Hubble flow, are superposed onto the cluster structure. While the virialized core is member-dominated, the contamination becomes critical at large cluster-centric distances, so that the density of nonmember galaxies increases significantly relative to the members.

3.3 Unveiling the Data Structure: The Exploratory Data Analysis

Before feeding the data into any ML algorithm, it is fundamental to perform an Exploratory Data Analysis (EDA). The EDA serves as the diagnostic step to understand the underlying structure of the dataset, verify the quality of the features, and assess their discriminative power regarding the target variable. In this work, the EDA focused on two main goals: analyzing the statistical properties and distributions of the whole dataset, and evaluating if these properties were sufficient for a robust classification.

An initial inspection of the primary observables in the mock catalogs, Right Ascension (α), Declination (δ), reveals that their direct use would severely limit the model's ability to generalize. A

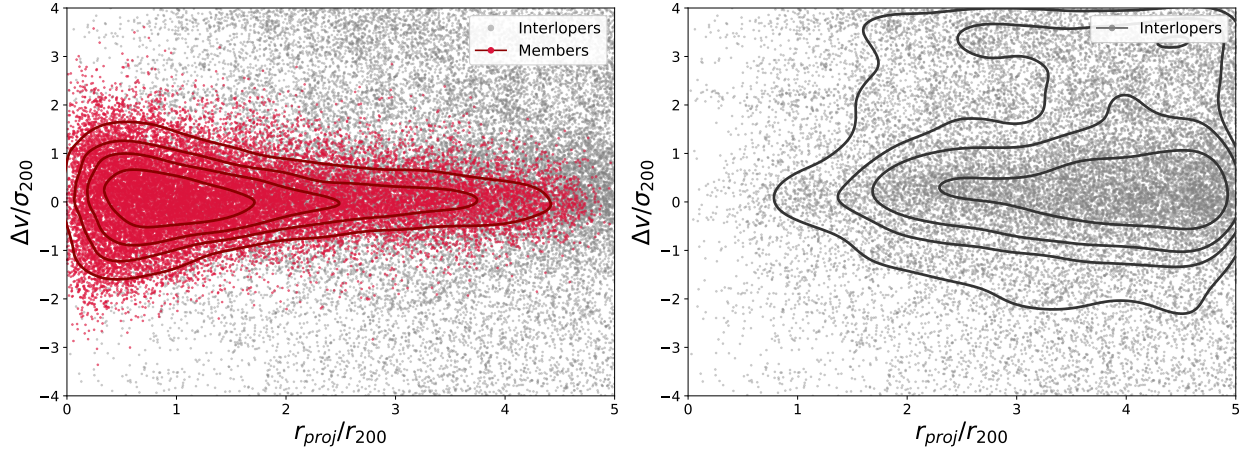


Figure 3.1: Projected position vs. velocity phase-space (PPS) diagrams for the 42 stacked mock clusters. The x axis corresponds to the normalized projected distance to the cluster center, whereas the y axis displays the normalized peculiar velocity to the cluster. *Left panel:* The distribution of all galaxies, showing Members (red) forming the characteristic "trumpet" shape and Interlopers (gray) forming a diffuse background. The solid contours trace the density levels of the member population, delineating their dynamical structure. *Right panel:* Distribution of Interlopers only, with contours indicating their density distribution. We note how contamination from projected galaxies is present across the entire PPS, becoming particularly severe at large radii ($r \gtrsim 2.5 r_{200}$).

classifier trained on raw sky coordinates would likely memorize the specific spatial locations of the training clusters rather than learning the physical rules of membership.

Consequently, the analysis confirms the necessity of transforming these raw observables into a physically meaningful, cluster-centric reference frame. To ensure the model learns dynamical patterns applicable to any cluster target regardless of its position in the sky, we must construct a new input space based on relative physical quantities.

3.4 Feature Engineering: Constructing the Input Space

Feature selection is one of the critical steps in the training process. We must transform the raw observational data (RA, DEC, z) into physically meaningful quantities that capture the dynamical state and environmental context of galaxies. Based on the EDA, we use five key features for the classification task. The calculation of these features requires defining the cluster center and redshift, which we assume to be known beforehand, as the target clusters have been previously identified by CHANCES (Sifón et al. 2025).

3.4.1 Dynamical Features

The primary dynamical discriminants for cluster membership are the galaxy's position and velocity relative to the cluster potential.

- **Projected Distance to the closest Cluster (r_{proj}):** The angular separation between the galaxy and the cluster center θ is converted to a physical projected distance using the angular diameter distance D_A , assuming a flat Λ CDM cosmology, with a Hubble parameter $H_0 =$

67.74 km/s⁻¹ Mpc⁻¹, a matter density parameter $\Omega_m = 0.3089$, and dark energy density $\Omega_\Lambda = 0.6911$, matching those used in the Uchuu simulation (Ishiyama et al. 2021). With the D_A at the cluster's redshift z_{cl} we obtain a metric in units of Mpc, independent of observational geometry:

$$r_{proj} = \theta \cdot D_A(z_{cl}) \quad (3.4)$$

Where θ is in radians. In order to distinguish galaxies located in regions where multiple clusters may overlap, we assign the distance to the geometrically closest cluster center.

- **Peculiar Velocity within the closest Cluster (v_{pec}):** The line-of-sight velocity difference between the galaxy and the cluster is often dominated by the cosmological redshift. We isolate the peculiar velocity component (v_{pec}) using the relativistic approximation (Davis and Scrimgeour 2014):

$$v_{pec} = c \cdot \frac{z_{gal} - z_{cl}}{1 + z_{cl}} \quad (3.5)$$

Where c is the speed of light constant and z_{gal} is the galaxy's spectroscopic redshift.

3.4.2 Environmental Features

Beyond global dynamics, the local environment of a galaxy provides critical information about its membership status, particularly for detecting substructures or overdensities.

- **Local Galaxy Density (Σ_{10}):** We estimate the local surface density using the distance to the N -th nearest neighbor (Fix and Hodges, 1951; Cover and Hart 1967) following the scheme described by Muldrew et al. (2012). We adopt $N = 10$ as a compromise between resolution and statistical robustness. Lower values (e.g., $N = 3$) are highly sensitive to Poissonian noise, while higher values (e.g., $N = 30$) tend to smooth out small-scale substructure relevant for infall regions, particularly for low-member clusters. The projected coordinates (x_{proj}, y_{proj}) for each galaxy were computed using the angular diameter distance at the cluster's redshift. The density is defined as:

$$\Sigma_{10} = \frac{10}{\pi d_{10}^2} \text{ [galaxies} \cdot \text{Mpc}^{-2}] \quad (3.6)$$

Where d_{10} is the projected distance to the 10th nearest neighbor in Mpc. High values of Σ_{10} are indicative of the dense cluster core or compact groups, whereas low values suggest field environments.

3.4.3 Intrinsic Galaxy Features

Finally, we include properties scaling with system mass or luminosity. Here, we purposely exclude galaxy properties related to evolution, such as color (e.g., $g - r$) or SFR. A primary scientific goal of CHANCES is to study how the environment transforms galaxies (e.g., quenching star formation) as they move from the infall region to the virialized core. If we were to include these properties, the algorithm would be biased to return the expected distributions, limiting the true potential of the algorithm. For instance, using color as a feature for membership classification would likely result in

the algorithm learning that "members are red" and "non-members are blue", preferentially selecting quenched galaxies and potentially excluding the population of blue, star-forming galaxies that are currently falling into the cluster. Consequently, we limit the intrinsic features to gravitational and luminosity proxies:

- **Associated Halo Virial Mass (M_{200}):** The virial mass of the galaxies' associated halo, in units of $M_{\odot}$. Including the host halo mass allows the model to adjust its decision boundaries according to the potential well size; for example, allowing higher peculiar velocities for massive clusters compared to less massive groups.
- **r-band Magnitude (m_r):** The apparent magnitude in the r-band as a proxy for galaxy luminosity.

3.4.4 Feature Correlation Analysis

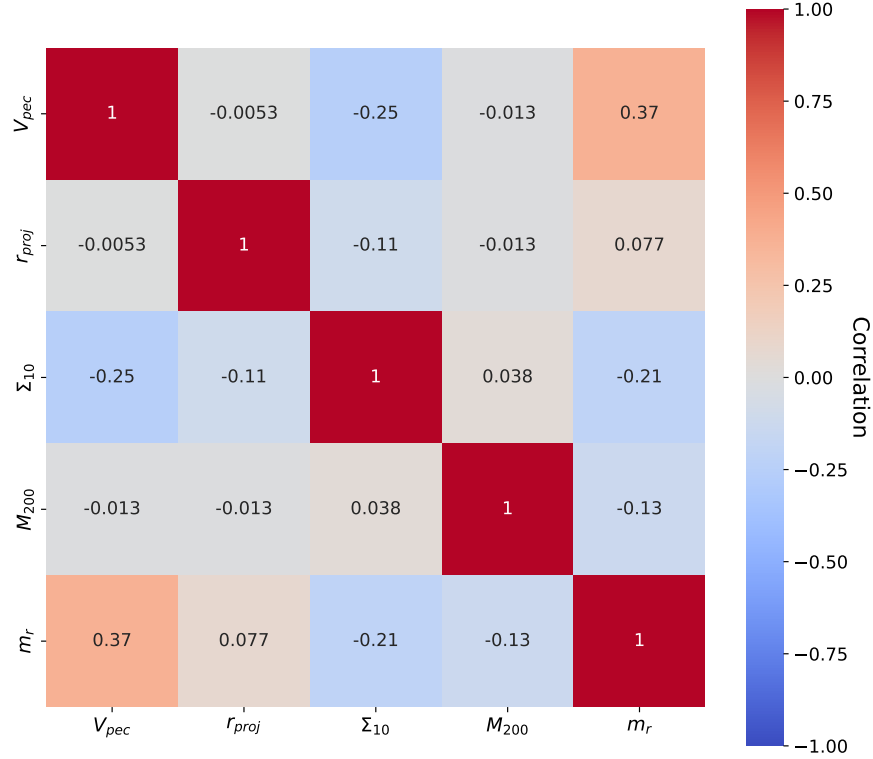


Figure 3.2: Pearson Correlation Matrix of the five input features selected for the Random Forest Classifier. The values indicate the linear correlation coefficient ρ between pairs of variables. The generally low values ($|\rho| \lesssim 0.4$) suggest that the features provide independent information to the model.

Once the feature definitions are clear, it is essential to assess the independence of the selected features. Highly correlated features (multicollinearity) can introduce redundancy, potentially biasing the interpretation of feature importance in tree-based algorithms like Random Forest (Drobnic et al. 2020).

To quantify the linear relationship between our five selected features, we computed the Pearson correlation coefficient (ρ) (Schober et al. 2018) for each pair of variables. The coefficient is defined as:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} \quad (3.7)$$

Where $\text{cov}(X,Y)$ is the covariance between features X and Y , and σ_α represents their respective standard deviations. The value of ρ ranges from -1 to 1, indicating linear and inverse linear correlations, while 0 indicates nonexistent linear correlation.

Figure 3.2 displays the resulting correlation matrix for our training dataset. The matrix reveals that the selected features exhibit low to, at most, moderate correlations. The highest observed correlation ($\rho \approx 0.37$) exists between m_r and v_{pec} , likely arising because brighter, more massive galaxies often reside centrally in massive halos where velocity dispersions are intrinsically higher. Other pairs, such as Σ_{10} and r_{proj} , show a weak negative correlation ($\rho \approx -0.11$), capturing the expected physical trend of decreasing density with distance while confirming that Σ_{10} provides distinct local topological information not captured solely by r_{proj} . Generally, the lack of strong correlations confirms that our feature space is well-constructed and that each variable provides complementary information.

3.5 The Machine Learning Algorithm: Random Forest

The classification of galaxy cluster members is formulated here as a binary supervised learning problem. We aim to learn a mapping function $f : X \rightarrow Y$, where X is the input feature vector described in Section 3.4 and $Y \in \{0,1\}$ represents the target class labels (0 for Interlopers, 1 for Members). Among the vast landscape of Machine Learning algorithms, we have selected the **Random Forest (RF)** classifier (Breiman 2001) for this task.

3.5.1 Decision Trees and Ensemble Learning

RF is an ensemble learning method constructed from multiple Decision Trees. A Decision Tree (DT; Quinlan 1986) is a non-parametric model that recursively partitions the feature space into branches, generating nodes. At each node, the algorithm selects the feature and the threshold that best splits the data into classes, aiming to maximize the Purity of the resulting child nodes. The quality of a split is typically measured by the Gini Impurity (Breiman et al. 1984; Tangirala 2020) or Entropy (Shannon 1948), also named Information Gain. We adopt the Gini Impurity as the splitting criterion due to its computational efficiency, as it avoids logarithmic calculations required by Entropy, while yielding comparable classification performance (Raileanu and Stoffel 2004). This metric is given by:

$$G = 1 - \sum_{k=1}^C p_k^2 \quad (3.8)$$

Where p_k is the fraction of samples that belongs to class k in a specific node. A $G = 0$ indicates a perfectly pure node, in other words, every sample belongs to the same class.

Nonetheless, individual DTs are prone to overfitting, as they can create overly complex structures that memorize the noise in the training set, leading to high variance and poor generalization

power. Furthermore, DTs are highly sensitive to minimal training set variations, which may result in unstable classifiers (Li and Belford 2001; Leiva et al. 2019; Amro et al. 2021).

3.5.2 Bagging and Feature Randomness

To mitigate the high variance of individual trees, RF employs *bagging* (bootstrap aggregating, Breiman 1996). Given a training set D of size n , bagging generates B new training sets $D_1, D_2, \dots, D_B$, each of size n , by sampling from D uniformly and with replacement. This means that some observations may be repeated in each D_b , while others are left out.

The RF algorithm trains a decision tree $\hat{f}_b$ on each bootstrap sample D_b . However, RF introduces a critical modification to standard bagging: *Feature Randomness*. When splitting a node during the construction of the tree, the split search is limited to a random subset of m features (typically $m \approx \sqrt{p}$, where p is the total number of features). This constraint prevents strong predictors from dominating the top splits across all trees, thereby ensuring that the individual estimators remain decorrelated, maximizing the variance reduction of the ensemble and enhancing generalization (Hastie et al. 2009).

In a classification problem, as illustrated in Figure 3.3, the final prediction $\hat{f}_{RF}$ for a new instance x is obtained by aggregating the predictions of the B individual trees via majority voting:

$$\hat{f}_{RF}(x) = \text{mode} [f_1(x), f_2(x), \dots, f_B(x)] \quad (3.9)$$

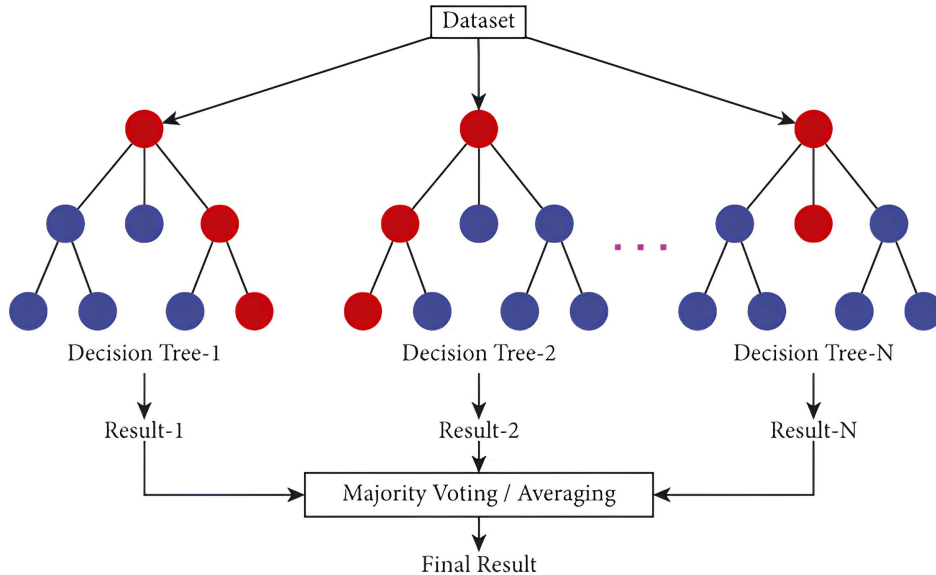


Figure 3.3: Schematic representation of the Random forest architecture. The model consists of an ensemble of independent decision trees, each trained on a different bootstrap sample of the original dataset, and the final classification is determined by the majority vote of the ensemble. Extracted from Khan et al. (2021)

We chose RF for this problem due to a variety of advantages suited for astrophysical tabular data:

- **High-dimensional non-linearity:** RF can identify complex, non-linear decision boundaries across the full N -dimensional feature space. This allows the algorithm to incorporate environmental cues (e.g., Σ_{10}) in order to disentangle overlapping populations that are indistinguishable in the $r_{proj} - v_{l.o.s}$ plane alone
- **Robustness to noise:** By leveraging a large amount of trees, RF reduces the variance associated with single DTs, making it a strong generalizer even in the presence of outliers in the mock catalogs.
- **Feature Importance:** Unlike "black-box" models like deep neural networks, RF provides intrinsic metrics to evaluate the importance of each feature in the classification decision through the Mean Decrease Impurity (MDI, Breiman et al. 1984), which is vital for physical interpretation. For a single tree b , the importance of a feature X_j is calculated by summing the decrease in Gini Impurity ΔG weighted by the proportion of samples reaching that node $p(t)$, across all nodes t where that feature is used for splitting:

$$\text{Imp}(X_j) = \frac{1}{B} \sum_{b=1}^B \sum_{t \in T_b: v(t) = X_j} p(t) \Delta G(t, X_j) \quad (3.10)$$

Where B is the number of trees, $p(t)$ is the fraction of samples in node t and ΔG the impurity reduction achieved by the split. This metric allows us to physically interpret which properties drive the membership classification.

- **Efficiency:** It handles tabular data efficiently due to parallelization and requires minimal data pre-processing (e.g., it is invariant to feature scaling), making it computationally accessible for large survey catalogs, a distinct advantage over algorithms such as Support Vector Machines, which typically scale worse with large sample sizes (Bottou and Lin 2007).

While Deep Learning models are powerful, they often require significantly larger datasets to outperform tree-based ensembles on tabular data and lack the immediate interpretability required to validate the physical basis of the classification (Shwartz-Ziv and Armon 2022). Given the structure nature of our catalog and the physical interpretability required, RF offers an optimal balance between complexity and performance.

3.6 The Training and Validation Strategies

A rigorous validation strategy is paramount to ensure the model's ability to generalize to new, unseen data. Standard validation techniques, such as random Test-Train splits or K-Fold cross-validation, are unsuitable for this problem because galaxies belonging to the same cluster are not independent instances; instead, they are correlated due to a shared dynamical environment and evolutionary history. Consequently, randomly splitting galaxies of a given cluster would lead to data leakage. This occurs when information from the validation set is exposed to the model during training, allowing it to "cheat". In this context, random splitting would grant the algorithm the necessary tools to memorize the specific phase space structure of a cluster, rather than learning the general physical rules of membership.

3.6.1 Leave-One-Group-Out Cross-Validation

To address this issue, we implemented a **Leave-One-Group-Out** (LOGO) cross-validation strategy¹. In this scheme, the dataset is partitioned into groups based on the physical association of galaxies within a cluster, rather than treating each galaxy as an independent and identically distributed sample. Our groups are defined by each of the individual clusters, yielding 42 unique groups.

For a 42 mock dataset, the LOGO strategy will iterate 42 times; for each iteration k , the model is trained on 41 out of the 42 clusters and tested exclusively on the single remaining cluster, and the final performance metrics are obtained by averaging the results across the entirety of folds. This approach has distinct advantages for our scientific goals:

- **True Generalization Assessment:** By testing on a completely unseen cluster, it provides a realistic estimate of the model's true generalization capabilities.
- **Robustness Analysis:** Since clusters vary in mass, dynamical state, and redshift, LOGO allows us to analyze the variance in performance across different cluster regimes.
- **Maximizing Training Data:** It ensures the model learns from the largest possible diversity of dynamical configurations in every iteration.

3.6.2 Model Optimization

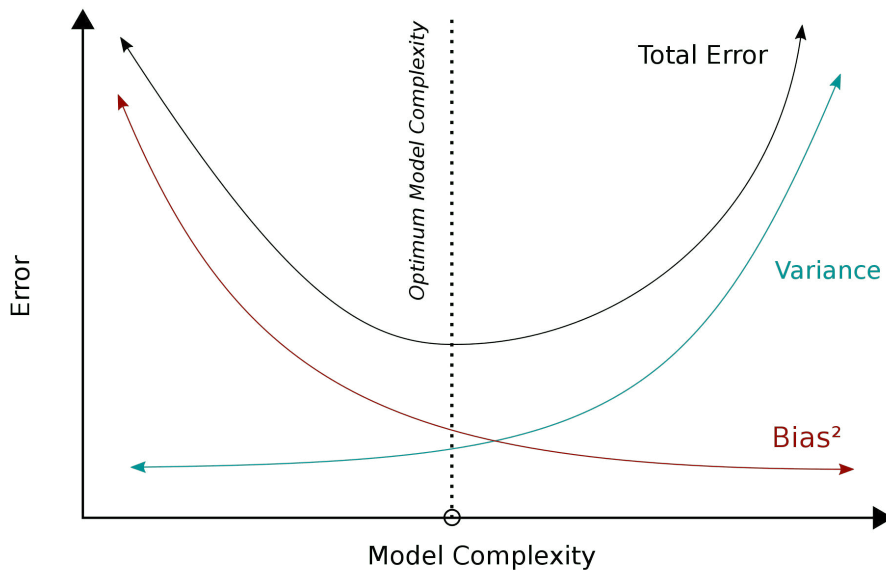


Figure 3.4: Representation of the Bias-Variance Trade-off. The plot illustrates how the total error (black curve)—sum of the variance (cyan) and the squared bias (red)—changes as a function of the model complexity. The vertical dotted line marks the optimum model complexity, where the error is minimized, representing an ideal balance. Extracted from <https://roysubhradip.hashnode.dev/understanding-the-bias-variance-trade-off>.

In supervised learning, the prediction error can be decomposed into two components:

¹https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.LeaveOneGroupOut.html

1. **Bias:** The error introduced by approximating a problem with a simplified model. High bias leads to underfitting, where the model fails to capture the underlying trends of the data.
2. **Variance:** The model's sensitivity to small fluctuations in the training set. High variance leads to overfitting, where the algorithm models the noise rather than the signal, as described for Decision Trees in Section 3.5.1.

As illustrated in Figure 3.4, as the model complexity increases, the bias decreases because it adapts better to the training data; however, the variance increases simultaneously, as the model becomes unstable and more sensitive to noise. In this regard, a key challenge to every supervised ML model is to find the optimum model complexity-where the total error reaches its minimum-optimizing the trade-off.

While the RF architecture naturally reduces variance through the averaging mechanism of bagging (Section 3.5.2), the complexity of the individual trees still plays a crucial role in the bias-variance dynamics. The trees' hyperparameters act as control knobs for this complexity; for instance, allowing trees to grow deeper decreases bias but increases variance, while constraining them does the opposite. Thus, identifying the precise configuration that minimizes the total error requires a systematic search strategy rather than manual trial and error.

To address this, and to maximize the model's generalization capability and performance, we performed hyperparameter tuning using **GridSearchCV**, nested within the cross-validation loop. As illustrated in Figure 3.5, this method explores all of the possible combinations for a given parameter grid and selects the best-performing model for the target metric.

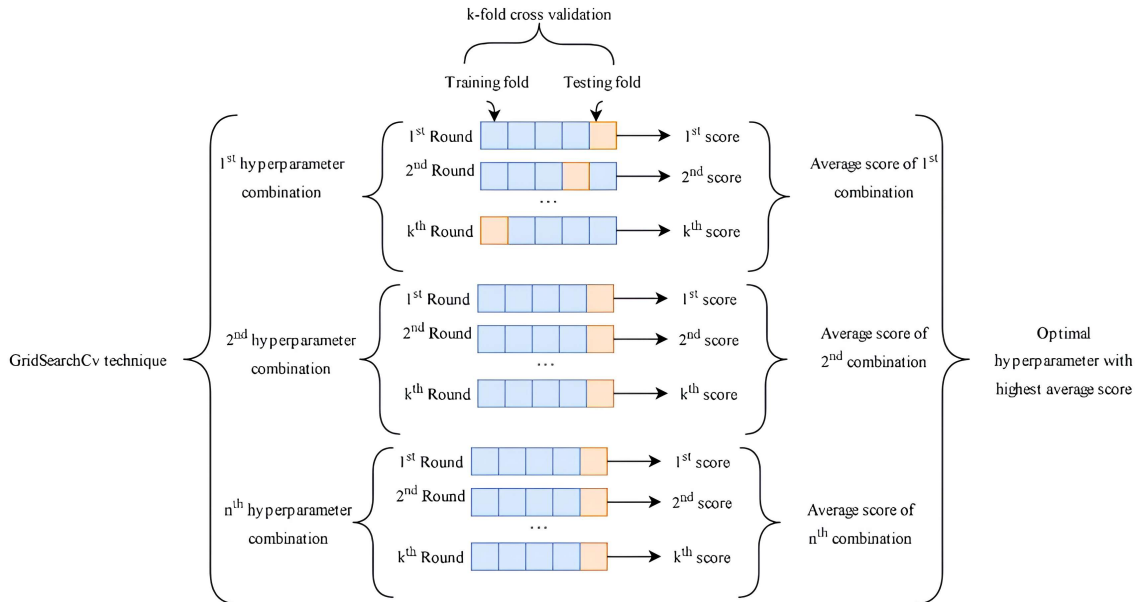


Figure 3.5: A schematic view of the GridSearchCV process. The diagram shows how the training set is divided into k folds to evaluate each hyperparameter combination. For every specific combination, the algorithm calculates an average score across all rounds. The combination that achieves the highest average score is selected as the optimal set of hyperparameters for the final model. Extracted from Kumar et al. 2024.

Specifically, we explored a grid of key parameters, chosen because of their influence on the bias-variance trade-off:

- **Number of trees (`n_estimators`):** While more trees generally lead to a more stable prediction by averaging out the variance, the performance gain typically reaches a asymptotic behavior after a certain threshold. We aimed to balance accuracy with computational efficiency.
- **Maximum Tree Depth and Complexity (`max_depth` & `max_leaf_nodes`):** These parameters control the maximum depth of the trees and the maximum number of terminal nodes, respectively. Unconstrained trees tend to grow until the tree leaves are pure, often memorizing noise in the training set and thus producing overfitting. By limiting the depth and the number of leaves, we force the model to learn more general patterns that hold true across different clusters.
- **Minimum Samples for Splitting (`min_samples_split`):** This parameter determines the minimum number of samples required to split an internal node. Higher values prevent the model from creating branches based on very few examples.
- **Impurity Reduction Threshold (`min_impurity_decrease`):** This sets a threshold for the split quality. As discussed in Section 3.5.1, the split at each node is made in order to maximize the purity of the child nodes, now, a node will be split only if the Impurity decrease induced is greater than the selected threshold, preventing the tree from making splits that provide marginal gains in separation while simultaneously introducing unnecessary model complexity.
- **Feature Selection per Split (`max_features`):** Restricting the number of features considered at each split forces the trees to de-correlate and discover patterns in weaker variables that might otherwise be overshadowed by the strong kinematic component. On the other hand, restricting the features in our low dimensional feature space ($N = 5$) may be a formula to exclude the most informative predictors from a split entirely, potentially leading to weak individual learners. We empirically determine that the benefit of accessing the full information content outweighs the gain from tree de-correlation.

The final optimal hyperparameters selected by the `GridSearchCV`, which provided the best balance between Purity and Completeness on the validation sets, are detailed in Table A.1.

3.6.3 Handling Class Imbalance

As highlighted in Chapter 2, our dataset exhibits a significant class imbalance ($\sim 11 : 1$). Here, it is important to note that this disparity is inherent to the problem itself; the vast background and foreground populations naturally outnumber the gravitationally bound members along the line of sight. Standard classifiers trained on such skewed data tend to be biased towards the majority class, often achieving high accuracy by simply classifying most of the majority class correctly while failing to recall the minority class (Levy et al. 2018). To treat this imbalance, we tested two complementary strategies:

- **Class Weighting (`class_weight`):** RF allows adjusting the cost function by assigning higher weights to the minority class. The weights are inversely proportional to class frequencies in the input data:

$$w_j = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_{\text{samples},j}} \quad (3.11)$$

Where w_j is the weight assigned to class j , n_{samples} is the total number of samples in the training set, n_{classes} is the number of classes, and $n_{\text{samples},j}$ is the total number of samples belonging to class j . This penalizes the model more severely for misclassifying a member than an interloper, effectively forcing the algorithm to focus its learning effort on the minority class examples.

- **Resampling techniques:** We explored methods that directly manipulate the dataset size. These techniques were implemented only on the training folds of the cross-validation to prevent data leakage.
 - **Synthetic Minority Over-sampling Technique (SMOTE):** SMOTE generates synthetic examples for the minority class by interpolating existing neighboring samples in feature space (Chawla et al. 2002). This technique aims to increase the size of the minority class, providing the classifier with a richer and more balanced representation of the member population. However, SMOTE can introduce artifacts or noise near the decision boundary, so a visual representation in the PPS is provided in Figure 3.6 to ensure that the data points created are indeed physically suitable. It is also important to note that this figure serves only as a representative illustration. Since the SMOTE step is nested within the training pipeline, distinct synthetic examples are generated for each cross-validation fold, meaning the specific synthetic points vary across iterations. Particularly, we over-sample the minority class until it reaches a 20% fraction of the majority class, enhancing the model’s sensitivity to members without discarding the realistic dominance of the interloper background.
 - **RandomUnderSampler (RUS):** RUS aims to balance the classes by randomly removing samples from the majority class². While simple and effective at class balancing and reducing training time, the main risk of RUS is discarding potentially useful information from the majority class, especially from interlopers located near the true decision boundary in the cluster outskirts. We under-sample the majority class until the minority class reaches 20% of the majority class.

To systematically evaluate the impact of these strategies, we defined three sampling scenarios: (1) **Original Data** (no sampling), (2) **SMOTE** (over-sampling), and (3) **RUS** (under-sampling). Since we also test each scenario both before and after **hyperparameter optimization**, this results in a total of **six different models** for final comparison.

3.7 Performance Assessment

To evaluate the RF model and establish a performance comparison between the six tested scenarios, we utilize a comprehensive set of metrics. To construct a catalog that minimizes interloper contamination while maximizing the recovery of genuine members, we prioritize metrics that quantify the trade-off between Purity and Completeness, particularly in the context of our highly imbalanced dataset.

²https://imbalanced-learn.org/stable/references/generated/imblearn.under_sampling.RandomUnderSampler.html

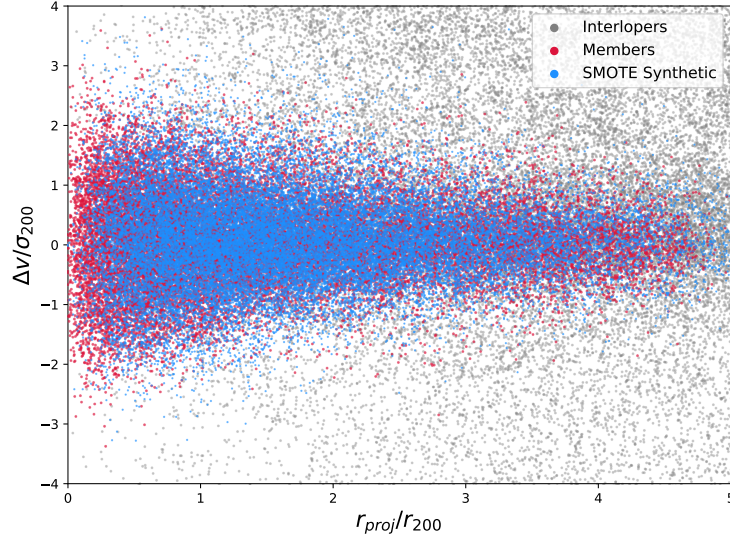


Figure 3.6: Distribution of the training set in the Projected Phase Space after applying SMOTE. The synthetic samples (blue) generated by interpolation respect the general trumpet shape of the original members (red), ensuring physically consistent training data.

3.7.1 Confusion Matrix and Classification Metrics

The foundation for evaluating a classification model in Machine Learning is the Confusion Matrix. This matrix summarizes the total number of correct and incorrect predictions made by the classifier by comparing the predicted class against the true class. In a binary classification scheme, the four key outcomes are:

- **True Positive (TP):** A positive instance correctly classified as a positive instance.
- **True Negative (TN):** A negative instance correctly classified as a negative instance.
- **False Positive (FP):** A negative instance incorrectly classified as a positive instance.
- **False Negative (FN):** A positive instance incorrectly classified as a negative instance.

Based on these outcomes, we define the key metrics relevant for our astrophysical application (Powers 2020):

- **Purity (Precision):** The fraction of predicted members that are true members. This quantifies the quality of the predicted member catalog, as it accounts for the contamination included.

$$\text{Purity} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3.12)$$

- **Completeness (Recall):** The fraction of true members correctly identified. This quantifies the ability of the model to recover all true members, including infalling and pre-processed galaxies.

$$\text{Completeness} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3.13)$$

- **F1-Score:** The harmonic mean between Purity and Completeness. It provides a single robust figure of merit that penalizes classifiers that perform well on one metric but poorly on the other, making it ideal for optimization on imbalanced datasets.

$$\text{F1-Score} = 2 \cdot \frac{\text{Purity} \cdot \text{Completeness}}{\text{Purity} + \text{Completeness}} \quad (3.14)$$

- **Accuracy:** The fraction of overall correct predictions:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}} \quad (3.15)$$

Due to the $\sim 11 : 1$ class imbalance, high accuracy can be misleading if the model simply classifies most galaxies as the majority class. Thus Accuracy is used only as a secondary, general performance indicator.

3.7.2 Decision Threshold and probabilistic metrics

The RF algorithm does not output a binary label directly; instead, it assigns a membership probability $P(y = 1|X)$ to each galaxy, based on the fraction of trees in the ensemble that voted for the Member class (Breiman (2001)). To convert this probability into a classification, a decision threshold P_{thresh} is applied.

By default, standard classifiers use $P_{thresh} = 0.5$; however, there is no universally "optimal" threshold; the choice depends on the specific scientific goals. In the context of galaxy clusters, a lower threshold increases Completeness, recovering more infalling galaxies at the cost of higher contamination, while a higher threshold increases Purity.

Due to the severe class imbalance, the model is inherently prone to False Positives when using a standard decision boundary. Consequently, a default threshold of 0.5 proves too lenient, allowing for an overwhelming contamination from interlopers that compromises the physical reliability of the catalog without offering a proportional gain in true member recovery. With this in mind, we explore the probability distribution of the predictions and adopt a conservative threshold of $P_{thresh} = 0.8$ for our final catalog. This optimized cut allows us to retrieve a sample that remains highly complete, crucial for characterizing the infalling population, while suppressing the excessive contamination that dominates at lower confidence levels, thus aligning our aims with state-of-the-art standards.

To evaluate the model's inherent ability to separate the classes independently of the chosen decision boundary, we use a threshold-independent metric such as the Receiver Operating Characteristic:

- **Receiver Operating Characteristic Area Under the Curve (ROC-AUC):** The ROC Curve (Hanley 1982; Fawcett 2006) plots the True Positive Rate (Completeness) against the False Positive Rate as the decision probability threshold is varied:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad , \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (3.16)$$

The AUC quantifies the probability that the classifier will rank a randomly chosen member higher than a randomly chosen interloper, giving an idea of the model's ability to distinguish between the classes.

3.7.3 Diagnostic Visualizations

Beyond numerical metrics, we employ graphical tools to diagnose the physical validity of the model:

- **Phase Space Mapping:** Predictions are mapped back onto the Projected Phase Space. By flagging each galaxy according to the four Confusion Matrix outcomes, we can visually identify where the algorithm successfully disentangles members from interlopers and, crucially, where classification errors occur. In addition, we inspect the distribution of predictions within the observational volume defined by RA, DEC, z . This allows us to verify that the selected members trace the cluster distribution on the sky and exhibit a coherent structure in velocity space, rather than getting associated with background structures.
- **Learning Curve:** This plot serves as a diagnostic tool to assess the stability and generalization capability of a model. By plotting a target metric (F1-Score in this work) for both the training set and the cross-validation set against the number of training examples used, we can analyze the convergence and gap between the two curves, helping to visualize the bias-variance trade-off (Hastie et al. 2009).
- **Feature Importance:** Finally, we analyze the ranking of features described in Section 3.5.2. Reviewing this hierarchy is vital to determine the main predictors, while simultaneously allowing us to interpret the decision-making process of the algorithm.

Chapter 4

Results

We present the quantitative evaluation of the Machine Learning models developed for the membership classification of galaxies in the CHANCES survey context. We first analyze the impact of class balancing strategies and hyperparameter optimization to select the best-performing model. Subsequently, we detail the performance of the selected classifier using global metrics, its behavior in the Projected Phase Space, and its bias-variance trade-off. Finally, we examine the radial dependence of Purity and Completeness, establishing a comparison with literature benchmarks, as well as analyze the feature importance hierarchy.

4.1 Model Comparison and Selection

To establish a baseline, we evaluated six scenarios based on the training data treatment: the raw dataset (Original Data), RandomUnderSampler (RUS), and Synthetic Minority Over-sampling Technique (SMOTE), each with their respective non-optimized and optimized variants. Table 4.1 summarizes the performance metrics.

Table 4.1: Comparative Performance of the Random Forest Models.

Model	Purity	Completeness	F1 - Score	Accuracy	ROC-AUC	Most Important Feature (Rate)
Original Data (Base)	0.87	0.45	0.59	0.95	0.98	v_{pec} (0.69)
Original Data (Optimized)	0.66	0.93	0.77	0.96	0.99	v_{pec} (0.96)
SMOTE (Base)	0.82	0.61	0.70	0.96	0.98	v_{pec} (0.64)
SMOTE (Optimized)	0.68	0.91	0.77	0.96	0.99	v_{pec} (0.96)
RUS (Base)	0.82	0.63	0.71	0.96	0.98	v_{pec} (0.65)
RUS (Optimized)	0.65	0.93	0.77	0.95	0.99	v_{pec} (0.96)

A direct comparison reveals that hyperparameter optimization significantly enhances performance across all metrics compared to the default `scikit-learn` configurations. While the Accuracy scores are consistently high (> 0.95) across all models, this metric is heavily biased by the class imbalance and primarily reflects the model's ability to classify the majority class correctly. Therefore, we focus on the F1-Score, Purity, and Completeness as reliable indicators.

While the non-optimized "Base" models achieve a misleadingly high Purity, they suffer from critically low Completeness (e.g., 0.45 for the Original Data model), meaning they fail to identify a significant portion of the true cluster members. The optimization process corrects this, boosting

Completeness at least a $\sim 30\%$ in every case—and even doubling the recovery rate for the Original Data model—effectively driving the F1-Score up. Furthermore, although not explicitly detailed in the Table 4.1, the non-optimized models exhibit training scores $\simeq 1.0$, a clear signature of overfitting. By constraining the model complexity (e.g., limiting tree depth), the optimized variants achieve a robust generalization, avoiding noise memorization and ensuring a more reliable classification of the target population.

All differences between the optimized versions of Original, SMOTE, and RUS are minimal. All three optimized strategies achieve high ROC-AUC scores (0.99), indicating excellent separability, as well as similar Purity, Completeness and F1 scores. This suggests that the internal handling of class imbalance provided by the RF algorithm—specifically through the use of `class_weight='balanced'` parameter—was sufficient to mitigate the bias towards the majority class. Consequently, external data manipulation techniques like SMOTE or RUS became redundant, as they did not provide a distinct advantage over the model’s intrinsic learning capabilities.

Given that the performance gain from resampling is negligible, and to avoid introducing potential artifacts (SMOTE) or reducing the statistical significance of the dataset (RUS), we identified the Original Data Optimized model as the primary candidate. To further validate this decision, we breakdown the confusion matrices.

4.2 Classification Performance

4.2.1 Confusion Matrices

The robustness of the optimized models and their advantage over default configurations are evidenced in the comparison of the normalized confusion matrices (Figure 4.1). Across the six tested scenarios, the optimized classifiers consistently achieve superior balance, with the selected model reaching a maximum True Positive Rate of 0.92 and a maximum True Negative Rate of 0.96. This balance is critical; while non-optimized models often yield a lower FP rate, they do so at the expense of a severe drop in Completeness (TPR), failing to recover a significant fraction of members due to a bias towards the majority class.

The contrast is deeply linked to the decision boundary $P_{thresh} = 0.8$ selected. For comparison purposes, the same threshold is used for every model, but due to the largely biased nature of non-optimized models, it prevents the recovery of a substantial fraction of genuine members, compromising the catalog’s utility for dynamical studies. However, it is worth noting that obtaining comparable results with optimized models would require manually tuning the threshold to lower values (e.g., ~ 0.28 for the non-optimized Base model, the F1 maximum), an approach that may not generalize well across different datasets. However, while hand-picking a specific P_{thresh} for each model could yield results tailored to the specific requirements of the CHANCES dataset, such an approach poses risks for model scalability.

If the scientific goal comprises ensuring the algorithm can generalize effectively to other low- z datasets without manual intervention, maintaining a consistent optimization pipeline is more effective than manually adjusting thresholds for every new case, although there is always the possibility for automating the threshold search based on specific scientific criteria, such as maximizing the F1-Score on the ROC curve or reaching a target Purity-to-Completeness ratio.

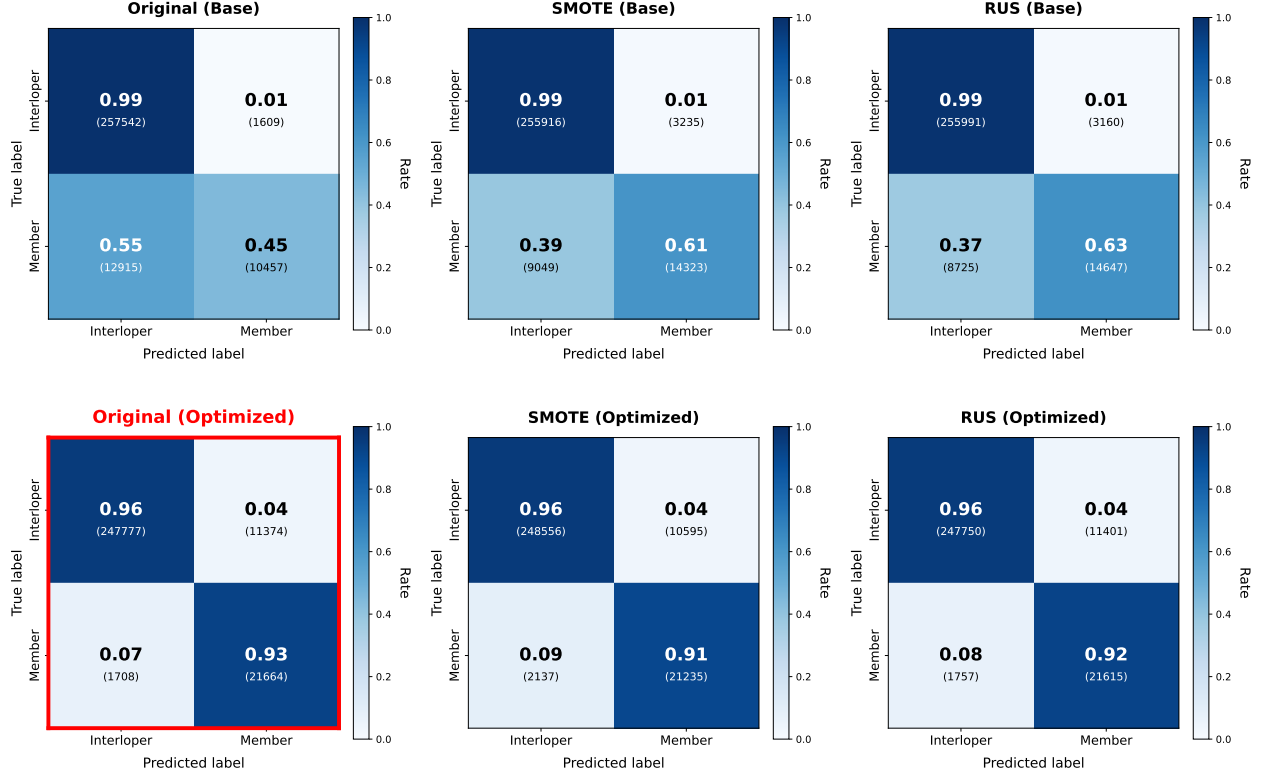


Figure 4.1: Normalized Confusion Matrices for the six Random Forest model configurations, comparing Base versus Optimized hyperparameters across three sampling scenarios: Original, SMOTE, RandomUnderSampler (from left to right). Rows correspond to true labels, while columns represent predicted labels. The diagonal elements indicate the recovery rate for each population, with the absolute galaxy counts shown in parentheses. The comparison highlights that optimized configurations significantly improve member Completeness to $\gtrsim 91\%$ and maintain a robust interloper rejection rate of 96% across all scenarios, effectively mitigating the majority-class bias present in the default models. The configuration selected for subsequent analysis is highlighted with a red border.

Considering the aforementioned, the confusion matrix analysis reinforces the idea that resampling techniques do not provide a significant performance advantage over the optimized model trained on the original distribution. Consequently, we select the **Original Data (Optimized)** model for the subsequent analysis. This decision prioritizes data integrity while achieving state-of-the-art classification performance.

4.3 Model results in Phase Space Analysis

Figure 4.2 (Left Panel) maps the model's predictions onto the Projected Phase Space. The distribution of TP correctly traces the cluster's dynamical structure, extending from the virialized core out to $\sim 4 r_{200}$ and, with lower density, reaching the $5 r_{200}$ limit. This confirms the model successfully identifies members even in the intermediate to low-density outskirts.

Classification errors follow a physically motivated pattern:

- **False Positives:** The contamination is predominantly located in the transition region between the infall zone and the field ($\sim 2.3 < r_{proj}/r_{200} < 5$), where the phase space density of members

overlaps with the Hubble flow. Minor yet still appreciable FP contamination exists in the central regions down to $\sim 0.5 r_{200}$.

- **False negatives:** Although rare (only ~ 1700 cases globally, see Figure 4.1), these missed members share the same phase space region as the FPs, mostly beyond $3 r_{200}$. This suggests that misclassifications are driven by the intrinsic dynamical degeneracy of the outskirts rather than model bias.

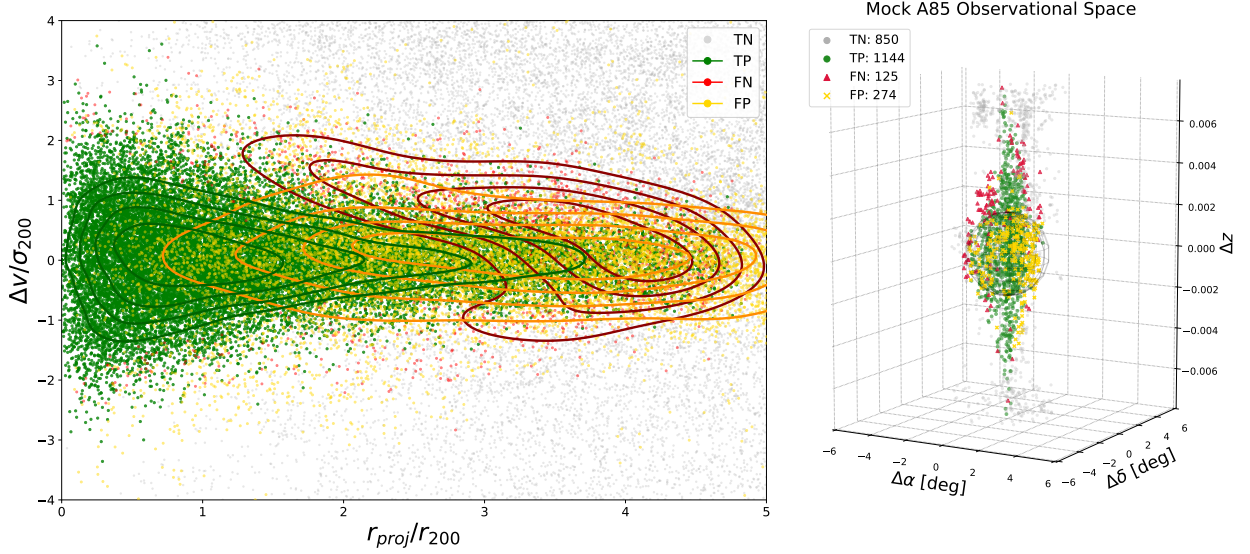


Figure 4.2: *Left Panel:* Diagnostic Projected Phase Space for the best Random Forest classifier. Axes are normalized by the virial radius r_{200} and velocity dispersion σ_{200} . Data points are color-coded according to the metrics of the confusion matrix: True Positives (green), True Negatives (gray), False Positives (yellow), False Negatives (red), including density contours enclosing down to 30% of each population. The model recovers the trumpet shape of the clusters, suffering mostly from FP misclassifications, with most errors occurring at the infalling region at $r_{proj}/r_{200} \gtrsim 2.5$. *Right Panel:* 3D distribution of galaxies for a representative cluster (Abell 85) in observational space. The displayed volume extends $3\times$ and $4.5\times$ the $5 r_{200}$ scale in the projected ($\Delta\alpha$, $\Delta\delta$) and line-of-sight (Δz) directions, respectively. The prominent elongation along the Δz axis illustrates the redshift distortion effects. The wire-frame sphere represents the cluster's projected physical scale ($5 r_{200}$) under pure Hubble flow. False Positives (yellow crosses) appear as interlopers distributed along the line of sight, while FN appear mostly outside of the $5 r_{200}$ sphere.

Furthermore, the 3D distribution shown in the right panel of Figure 4.2 offers a visual inspection of the classification performance in observational space ($\Delta\alpha$, $\Delta\delta$, Δz) using Abell 85 as a representative cluster. Unlike a real-space map, this view reveals the severe impact of redshift-space distortions. The TP do not form a compact sphere but define a distinct elongated structure stretching along the line of sight well beyond the physical virial boundaries. This elongation is the direct manifestation of the "Fingers of God" (FOG) effect (Jackson 1972), where the high virial velocities of members are observationally interpreted as spatial depth. Crucially, the FP align with this kinematic cylinder, confirming that contamination is driven by background and foreground interlopers whose peculiar velocities shift their observed redshifts into the cluster's dynamical range, making them indistinguishable from the elongated member distribution in projection. The FN are predominantly found outside the $5 r_{200}$ sphere; by cross-referencing their location with the PPS analysis (right panel), we identify these objects as tracers of the infall population; unlike the high-velocity virialized members,

these galaxies lack a strong kinematic signature, causing their signal to overlap with the ambient field flow, leading to some misclassifications.

4.4 Radial Dependence of Purity and Completeness

To assess the reliability of the catalog as a function of the cluster-centric distance, we analyzed the radial profiles of Purity and Completeness in Figure 4.3.

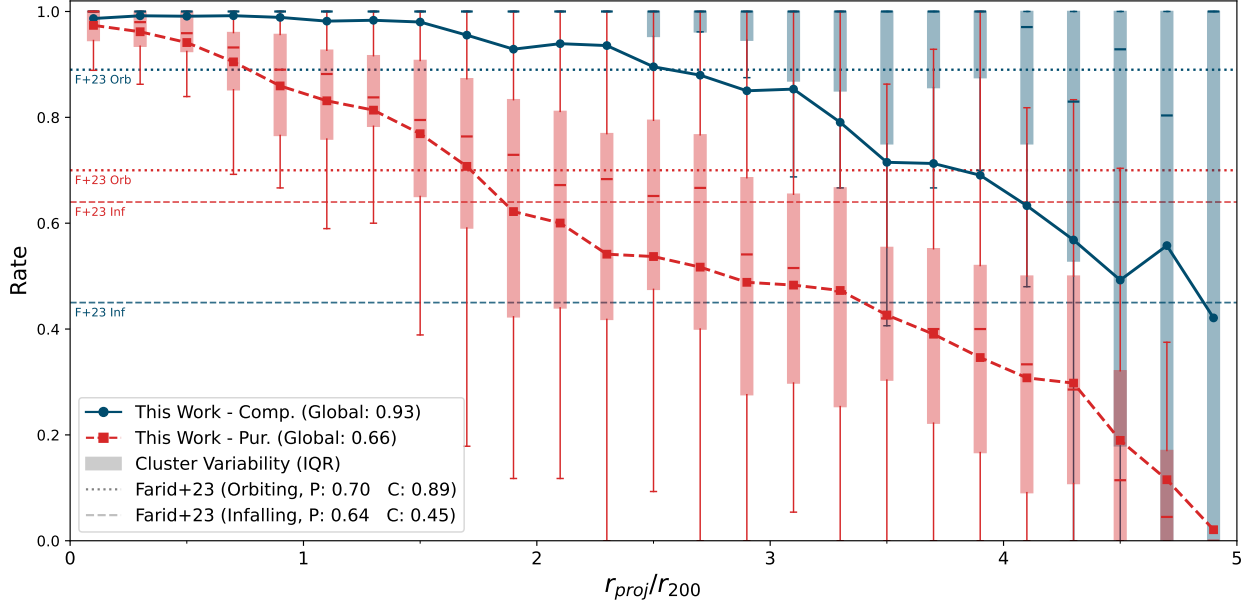


Figure 4.3: Radial profiles of Purity (red squares) and Completeness (blue circles) as a function of projected cluster-centric distance, calculated in radial bins of width $0.2 r_{proj}/r_{200}$. The global metrics are shown with solid lines/markers, while the cluster-to-cluster variability (do not confuse with ML variance) is depicted by the box-plots (interquartile range). Horizontal dotted lines represent the benchmark performance from Farid et al. (2023) for Orbiting and Infalling populations. The results show that the model maintains high Completeness out to large radii, while Purity exhibits a linear decline consistent with the increasing interloper density in the outskirts.

- **Completeness:** The model exhibits remarkable stability, with a perfect global Completeness of ~ 1.0 up to $\sim 1.5 r_{200}$. The decline is gradual, remaining above 0.4 even at the extreme limit of $5 r_{200}$. Notably, for at least 50% of the clusters (blue median lines in the box-plots), the Completeness remains close to 1.0 up to $4 r_{200}$, highlighting the model's effectiveness in recovering the infall population.
- **Purity:** The precision shows a stronger radial dependence, decreasing almost linearly with radius, going from 1.0 at $5 r_{200}$ to near 0 at $5 r_{200}$, yet holding up to 0.60 at just over $2 r_{200}$. This behavior is expected, as the ratio of interlopers to members increases exponentially towards the outskirts, making the task increasingly challenging.

Additionally, the box-plots reveal that the variance between individual clusters increases significantly with radius, spanning the entire y-axis at large cluster-centric distances. This indicates a loss

of stability in the outskirts, which, consistent with the mean trends, manifests at smaller radii for Purity when compared to Completeness.

We also establish a comparison with the C^2 -GAME framework (see Table 1. in Farid et al. 2023). Since that work distinguishes between Orbiting and Infalling populations, we included benchmarks for their best-performing deterministic models for both the Orbiting and Infalling populations. Our global Completeness of 0.92 significantly outperforms their reported recall for the Infalling population (~ 0.45) and it remains comparable to the Orbiting population (~ 0.89), verifying that our binary classification strategy successfully recovers the elusive infall component. Regarding Purity, our global 0.65 result remains competitive with their Orbiting and Infalling benchmarks of ~ 0.70 and ~ 0.64 respectively.

4.5 Bias-Variance Trade-off with the Learning Curve

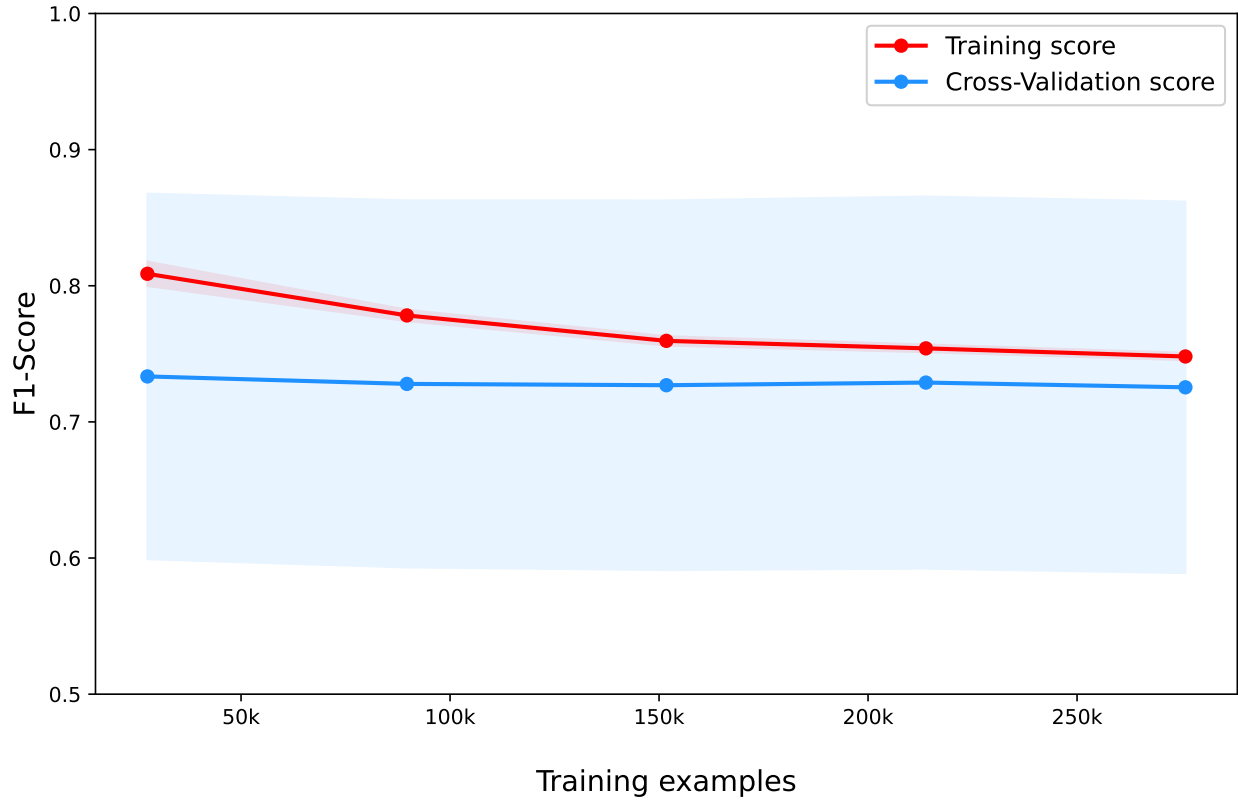


Figure 4.4: Learning Curve for the best Random Forest model. The plot shows the evolution of the F1-Score as a function of the number of training examples. The red line represents the training score, while the blue line denotes the cross-validation score obtained via the LOGO scheme. The narrow and stable gap between both curves indicates that the model is robust against overfitting and is a good generalizer for new data. An early asymptotic behavior in the training score points towards the fact that the model successfully captures the underlying dynamical patterns of cluster membership with a low count of training examples.

The learning curve (Figure 4.4) demonstrates the model’s convergence and stability. The gap between the training score and the cross-validation score is initially small (~ 0.07) and narrows

further as the number of training examples increases, indicating no significant overfitting. Thus, the model generalizes well to unseen data.

A notable feature is the almost flat trajectory of both the training and cross-validation score, which suggests that the algorithm quickly captures the underlying complexity of the problem. A cross-validation score stabilized at ~ 0.75 is coherent with the expected ceiling given the intrinsic noise of the dataset. A noticeable standard deviation for the CV score of $\sigma \sim 0.15$ is attributed to the LOGO validation strategy; since each fold evaluates a distinct cluster with unique dynamical substructures and merger histories, this variance reflects the natural physical diversity of galaxy clusters rather than model instability.

4.6 ROC Curve and Discrimination Capability

Figure 4.5 displays the Receiving Operating Characteristic, which exhibits an Area Under the Curve of 0.99, demonstrating a near-perfect intrinsic capability to distinguish members from interlopers. While this high value can be somewhat influenced by the ease of classifying the distinct background population, the curve confirms that the probabilistic ranking is robust.

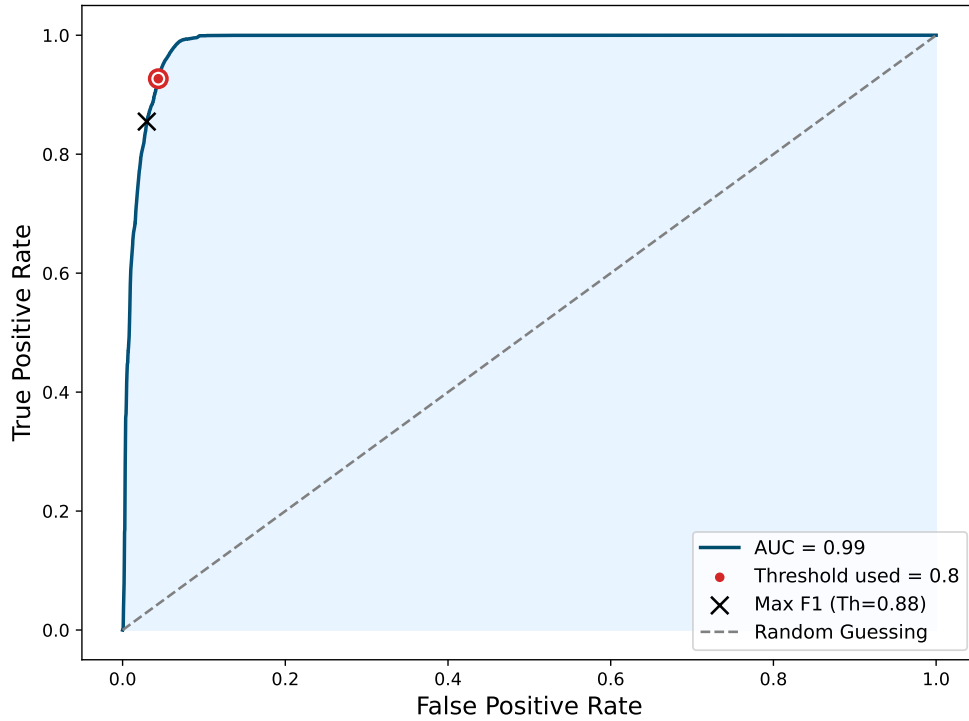


Figure 4.5: Receiver Operating Characteristic curve for the best Random Forest model. The blue line represents the trade-off between the True Positive Rate and the False Positive Rate as the decision threshold varies. The Area Under the Curve of 0.99 indicates a near-perfect ability of the classifier to distinguish between members and interlopers. The red dot marks the operational threshold adopted in this work ($P_{thresh} = 0.80$), which sits near the "elbow" of the curve, while the black cross indicates the threshold that mathematically maximizes the F1-Score ($P_{thresh} = 0.87$). The dashed diagonal line represents the performance of a random classifier.

Analysis of the curve identifies the mathematical optimum for the F1-Score at a threshold of 0.87, maximizing the Purity-Completeness trade-off. Nonetheless, adopting this threshold would yield a Purity of 0.72 and Completeness of 0.86. For our scientific goals, maximizing the recovery of infalling galaxies while minding not to introduce too much FP contamination is a priority. Hence, the operational threshold $P_{thresh} = 0.8$ determined offers a more strategic trade-off, maintaining high Purity for galaxy cluster membership standards while pushing Completeness to 0.92.

4.7 Feature Importance

Finally, the interpretation of the model is grounded in the feature importance analysis (Figure 4.6). The results unequivocally show that the peculiar velocity to the nearest cluster v_{pec} is the dominant predictor, with a relative importance of ~ 0.96 . The projected distance to the closest cluster r_{proj} plays a minor role (0.037), while other properties like local density or magnitude have negligible impact.

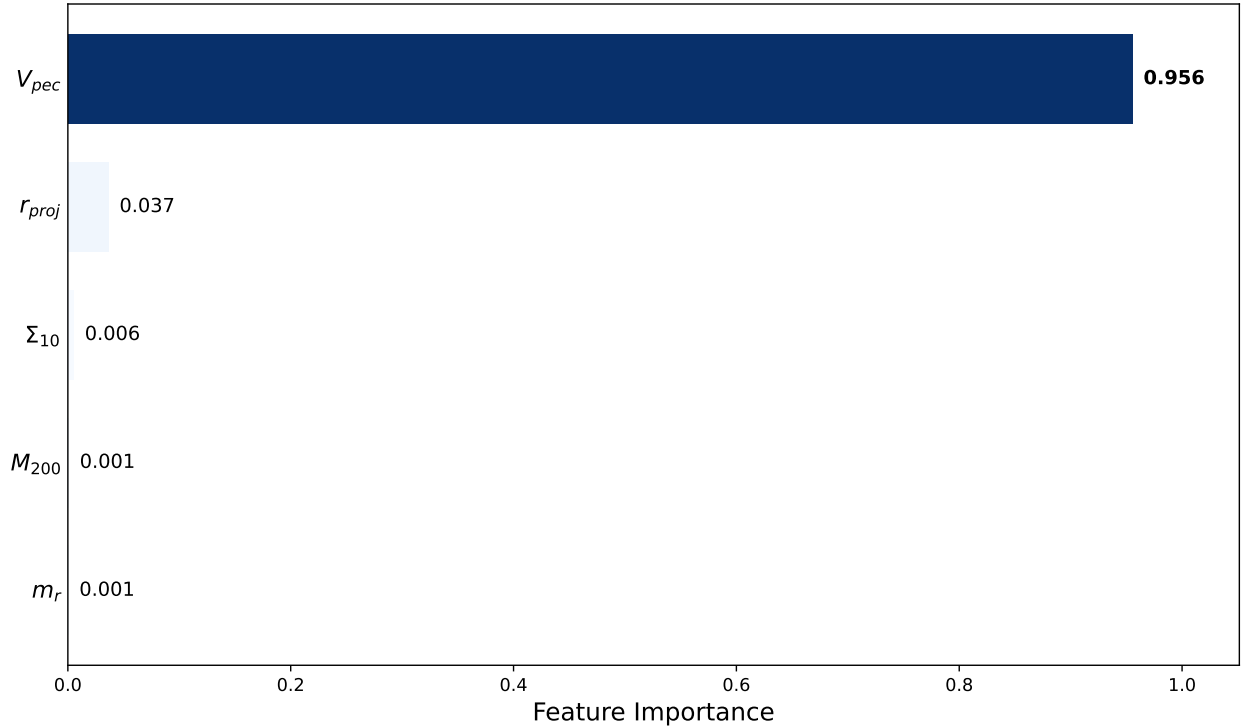


Figure 4.6: Random Forest Feature Importance for the best model. The bars illustrate the the relative contribution of each input variable to the decision-making process. The peculiar velocity relative to the cluster v_{pec} is the overwhelmingly dominant feature, accounting for 95.6% of the importance, while other features show low to negligible impact.

This hierarchy indicates that the Random Forest essentially behaves as a dynamical filter, leveraging velocity information to define the membership probability, while r_{proj} acts as a secondary constraint to refine the decision boundaries in very particular scenarios. The physical implications of this behavior will be addressed in the Discussion section.

It is worth noting that for non-optimized models, while v_{pec} remains the primary feature, we observe a higher relative incidence of secondary variables, particularly r_{proj} (see Table 4.1). This slight shift suggests that without hyperparameter tuning, the model relies more heavily on geometric constraints to compensate for the lack of optimal decision boundaries in the velocity dimension.

Chapter 5

Discussion

We interpret the results presented in Chapter 4, focusing on the physical implications of the model's behavior. We address the dominance of kinematic features, the intrinsic limitations imposed by projection effects, and the trade-offs involved in feature selection for galaxy cluster membership.

5.1 The Random Forest as a Dynamical Filter

The performance evaluation indicates that the optimized Random Forest model is highly robust, achieving a global Completeness of ~ 0.93 . This indicates that the classifier successfully recovers the vast majority of cluster members, not only in the virialized core but extending effectively out to $5 r_{200}$. A key indicator of the model's success is its generalization capability; the stability of the metrics suggests that the model has avoided overfitting, learning the underlying dynamical structure of the clusters rather than memorizing the noise of the training set.

The most striking result from the feature importance analysis (Figure 4.6) is the overwhelming dominance of the peculiar velocity v_{pec} , which accounts for 96% of the decision-making power. This ranking suggests that the algorithm has essentially learned to construct a dynamical barrier in phase space. By prioritizing v_{pec} and adjusting with the projected distance r_{proj} , the model autonomously reconstructs the characteristic trumpet shape of the clusters in the Projected Phase Space, as seen in Figure 4.2. It correctly identifies that the velocity dispersion of members must decay with radius, effectively reproducing the cluster's escape velocity profile. This behavior mimics classical dynamical methods, such as the caustic technique (Diaferio and Geller 1997; Diaferio 1999). However, distinct from the caustic method which derives the escape velocity profile that often require careful calibration, our model learns this boundary empirically from the simulation's ground truth through a supervised learning framework, allowing for a fully automated and homogeneous application across large datasets, bypassing the potential biases introduced by the manual or heuristic fine-tuning inherent to traditional phase-space estimators.

However, this heavy reliance on kinematics reveals a fundamental limitation: the model serves as an excellent "geometric" separator of the PPS, distinguishing the virialized structure from the field, but it struggles to differentiate classes within the caustic boundaries. Since the model treats the problem almost exclusively through a 1D kinematic lens, any interloper that falls within the cluster's velocity range is almost indistinguishable from a true member without additional information.

5.2 Projection Effects and Redshift Distortions

The core complexity of this classification problem lies not in the algorithm’s capacity, but in the degeneracy of the observational data. This is visually confirmed by the observational 3D space analysis of Abell 85 in the right panel of Figure 4.2. Here, the TP do not form a compact sphere but define a distinct cigar-shaped structure stretching along the line of sight, a direct manifestation of the FOG effect mentioned in Section 4.3, where the high virial velocities of members are observationally interpreted as spatial depth.

In observational astrophysics, we lack access to the 3D positions; instead, we use z as a proxy for the distance. However, z is degenerate: it traces both the cosmic expansion (Hubble flow) as well as the peculiar velocity of the galaxy due to local gravitational interactions.

$$v_{obs} \approx H_0 d + v_{pec} \quad (5.1)$$

This creates a crucial ambiguity that affects the core and the outskirts differently:

1. **Virialized Regime (Fingers of God):** In the cluster core, members move with high random velocities, often exceeding $\sigma_v > 800$ km/s. When observed in redshift space, these high velocities are interpreted as large distances along the line of sight, stretching the cluster into an elongated structure that points towards the observer (see right panel on Figure 4.2). This stretching causes the cluster members to overlap in redshift space with foreground and background galaxies that are physically distant but coincident in projection (Jackson 1972).
2. **Infall Regime (Hubble Flow Cancellation):** In the outskirts, the dynamics are dominated by coherent infall rather than random motion. Here, a field galaxy situated well beyond the cluster but with a negative v_{pec} (pointing towards us) can cancel out the Hubble expansion difference. Consequently, its observed redshift v_{obs} falls precisely within the cluster’s dynamical range, creating a "kinematic camouflage" indistinguishable in projection (Kaiser 1987; Diaferio 1999).

The model’s errors in the outskirt are then driven by this second degeneracy. The FP identified in the $\sim 2.3 - 5 r_{200}$ region align with the kinematic cylinder of the cluster, corresponding to field galaxies whose projected kinematics mimic the cluster’s infall caustics.

A way to quantify this limitation is to consider a significant yet realistic cluster velocity dispersion of $\sigma_v \sim 1000$ km/s, which translates to a Hubble flow distance of ~ 14 Mpc h^{-1} . This means that any galaxy within a cylinder of ~ 28 Mpc h^{-1} length along the line of sight is a potential interloper if its peculiar velocity aligns unfavorably.

The spatial distribution of FN in the 3D observational plot offers insight into their misclassification. Unlike the core members that are elongated into the finger structure, the FN are predominantly located outside the reference sphere. This visual placement suggests that these members possess low line-of-sight peculiar velocities, making them kinematically alike to the surrounding Hubble flow of the field. Without additional information to break this degeneracy, no purely kinematic hyperplane can perfectly separate these overlapping populations in projection.

5.3 Algorithmic vs. Physical Limitations

The analysis of the learning curve (Figure 4.4) reinforces the hypothesis that the limitations are physical rather than algorithmic. The validation score saturates quickly, flattening the trajectory as the training size increases. While this rapid "plateau" behavior is often considered as underfitting (i.e., the model being too simple), given the RF's intrinsic capacity for high complexity coupled with an optimization process that fine-tunes the hyperparameters to fully exploit this capacity, this trend actually indicates that the model has successfully extracted most of the available information from the provided feature space. Furthermore, this saturation demonstrates that the dataset size is not a limiting factor; the current training sample is sufficient and simply adding more data would accomplish nothing.

This hypothesis is further supported by the outcome of the resampling experiments conducted to address the inherent class imbalance in the outskirts, including random under-sampling of the majority class utilizing RUS and synthetic over-sampling of the minority class using SMOTE. Remarkably, none of these strategies yielded statistically significant improvements in the F1-Score compared to the standard model. This insensitivity to class distribution provides definitive evidence that the classification barrier is not statistical (i.e., caused by a lack of examples or imbalance) but fundamental: the overlapping populations are topologically inseparable in the projected feature space. Thus, the performance saturation is not a failure of the algorithm to learn from the minority class, but rather a manifestation of the hard physical ceiling imposed by projection effects. This validates our choice to retain the standard class-weighted model, as further algorithmic tuning yields diminishing returns.

With this in mind, the F1-Score ceiling is not a failure of the Machine Learning, but a quantification of the intrinsic noise caused by projection effects. To improve this metric, one would strictly need to expand the feature space. However, incorporating galaxy properties such as color, stellar mass, or SFR introduces significant challenges. First, it creates a circularity bias: if the classifier learns that members are "red and dead", it will systematically exclude the faint star-forming population—crucial for studying pre-processing—that we aim to recover.

Second, unlike direct observables, some of these physical parameters are model-dependent quantities, derived through calibrations that rely on assumed Initial Mass Functions (IMF), Stellar Population Synthesis (SPS) models, or dust attenuation laws.

Accordingly, we consider the proposed model to represent an optimal compromise. It maximizes the recovery of members with Completeness 0.93 purely based on dynamics, while the resulting Purity of 0.66 reflects the acceptance of a controlled level of contamination. This trade-off is not an algorithmic defect, but rather an unavoidable physical consequence of observing a 3D Universe in 2D.

Chapter 6

Conclusions

We have developed and optimized a Machine Learning framework to tackle one of the most persistent challenges in extragalactic astronomy: the identification of galaxy cluster members out to large cluster-centric distances. By leveraging the high-quality mock catalogs from the upcoming CHANCES survey—which will utilize the 4MOST facility to obtain extensive spectroscopy out to $5\ r_{200}$, enabling the characterization of galaxy pre-processing prior to cluster infall into the virial region—we trained a Random Forest classifier to distinguish between members and interlopers using almost exclusively velocity (v_{pec}) vs. projected position (r_{proj}) information. Based on the analysis of the model’s performance, the physical interpretation of the feature space, and the constraints imposed by observational data, we summarize our main findings below.

We have demonstrated that a purely kinematic Machine Learning approach is a viable strategy to recover cluster members beyond the virialized core. The optimized Random Forest model acts as a dynamical filter, achieving a high global Completeness of ~ 0.93 , confirming that the algorithm successfully traces the member distribution seamlessly in the core but with radial decrease that is increasingly challenging deep in the outskirts.

The cleaning efficiency, nevertheless, is strongly dependent on cluster-centric distance. While highly effective in the inner regions, the Purity exhibits a steep radial decline, approaching zero at the survey limit of $5\ r_{200}$. This behavior reflects the overwhelming dominance of interlopers in the outskirts, where the projected phase space density of the cluster becomes indistinguishable from the background. Consequently, rather than generating a pristine catalog at large radii, the model prioritizes the recovery of infalling members—minimizing missing detections—at the inevitable cost of including significant contamination in the outermost region.

Crucially, the model was deliberately restricted to use only projected position and velocity features, excluding galaxy properties such as color or mass to avoid biasing future scientific inquiries. Despite this self-imposed constraint, the algorithm reached a performance "plateau", extracting virtually all available information from the phase space. The convergence of the learning curves indicates that the model is well-optimized and free from overfitting, providing a generalized solution applicable to the diverse cluster morphologies expected in the CHANCES survey.

A central finding of this work is that the limiting factors in membership determination are fundamentally physical, not algorithmic. We identified that the primary source of contamination arises from the intrinsic degeneracy of projecting a 6D phase space onto 2D observational coordinates. While the overwhelming dominance of v_{pec} (96% importance) allows the model to identify

the trumpet shape of the escape velocity profile, effectively cleaning the virialized structure, our 3D observational analysis confirms that the model hits a "hard ceiling" in the infall regime.

Neither increasing the training size (as indicated by the learning curve in Figure 4.4) nor altering the class balance via over-sampling or under-sampling techniques produced any statistical improvement, confirming that the overlapping populations are almost inseparable in the current feature space. Consequently, we selected the standard Random Forest model as the optimal compromise, maximizing Completeness (~ 0.93) while accepting a Purity of ~ 0.66 . Improving this metric significantly without further feature space engineering is unfeasible due to the intrinsic overlap of these populations in the projected phase space.

Ultimately, our results validate the use of supervised Machine Learning as a powerful alternative to classical dynamical methods. The Random Forest autonomously reconstructed the non-linear boundaries of the cluster in phase space, mimicking the results of the caustic technique but with a significant methodological advantage. Unlike classical methods, which often rely on heuristic assumptions or the manual calibration of kernel parameters, our ML approach is purely data-driven. This ensures a homogeneous and automated application across large datasets, making it particularly suitable for next generation surveys, where the consistent treatment of thousands of clusters is required.

Chapter 7

Future Work

The Machine Learning framework developed in this thesis represents a robust first step towards automated membership classification in high cluster-centric regimes. Nevertheless, the analysis has also highlighted intrinsic limitations imposed by the current feature space and the projection effects. Based on our findings, we identify several key avenues for future research to enhance the precision and applicability of this method within the CHANCES survey and beyond.

The most immediate and critical extension of this work is the application of the trained Random Forest model to the upcoming spectroscopic data from the CHANCES Low- z survey. Since the model has been trained on mock catalogs specifically designed to replicate the survey’s selection function and target properties, the transition to real observations is expected to be reasonably good.

Beyond the binary classification presented here, future iterations should move towards a fully probabilistic framework. While the Random Forest natively outputs probability estimates, this work utilized a deterministic threshold for evaluation. Adopting a probabilistic approach would allow subsequent scientific analyses to weight galaxies according to their membership likelihood, effectively propagating the classification uncertainty into mass profiles or luminosity functions. Furthermore, the classification scheme could be refined to distinguish between dynamical sub-populations, specifically separating virialized (orbiting) members from the infalling population, similarly to the one presented by Farid et al. (2023). Successfully disentangling these components would allow us to map the accretion history of clusters dynamically, isolating the galaxies entering the potential well for the first time from those already processed by the cluster environment.

Regarding the methodological limitations identified, particularly the physical ceiling caused by projection effects, our results suggest that simply swapping the Machine Learning architecture (e.g., Deep Learning Neural Networks) while maintaining the same tabular input would not yield significant improvement. Consequently, if the analysis continues to rely on tabular data, priority must be placed on a more advanced feature engineering, rather than algorithmic complexity. Future work should focus on feeding the model with richer information to better contextualize the galaxy’s environment and cluster’s dynamical state specifics.

In parallel, specific pre-processing steps must be implemented to mitigate the FOG distortions. A statistical virial compression technique, similar to the one described in Shi et al. (2016)—where the member distribution is modeled with a NFW profile (Navarro et al. 1997) and line-of-sight positions are reassigned to statistically suppress the high velocity dispersion—can counteract the elongation

into a more compact distribution. By applying this technique before training, we can shrink the "kinematic cylinder", reducing the overlap with the Hubble flow and simplifying the decision boundary for the classifier.

In the longer term, and particularly as data volume increases with the new generation surveys, exploring relational Deep Learning architectures represents a viable experimental path. Graph Neural Networks (GNNs) could model the cluster as a connected graph rather than a collection of independent data points. GNNs offer a distinct advantage over environmental metrics that reduce the environment to a single scalar value; instead, they employ mechanisms to aggregate the full feature vectors of neighboring instances, allowing the model to learn from the specific properties of the local population (Zhou et al. 2018). However, applying GNNs to projected data presents its own challenges; if the input features remain degenerate, the graph construction (based on proximity) may inadvertently link interlopers to true members, reinforcing false positives. Thus, GNNs should be viewed as a tool to exploit local correlations only after the primary projection effects have been mitigated via the feature engineering outlined above.

Finally, to grant an extra layer of robustness to the pipeline, systematically investigating the variance in model performance observed across different mock clusters is key. Future studies should correlate classification metrics with global cluster properties, such as halo mass, dynamical state, and formation time, to understand in which regimes the model performs best. Additionally, verifying the model's generalization capability by testing it on independent external Low- z datasets (e.g., spectroscopic catalogs from SDSS or GAMA) would serve as the ultimate validation of the universal applicability of this Machine Learning framework.

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Appendix A: Hyperparameter Optimization Settings

Table A.1: Final hyperparameters of the optimized Random Forest classifier. Parameters not listed here were kept at their default `scikit-learn` values.

Hyperparameter	Value	Description
<code>n_estimators</code>	300	Number of trees in the forest.
<code>max_depth</code>	20	Maximum depth of the tree.
<code>max_leaf_nodes</code>	80	Maximum number of leaf nodes.
<code>min_samples_split</code>	50	Minimum samples required to split a node.
<code>min_impurity_decrease</code>	0	Minimum decrease in impurity required to split a node.
<code>class_weight</code>	<i>Balanced</i>	Class weighting.
<code>max_features</code>	None	Features considered at each split.
<code>criterion</code>	Gini	Function to measure quality of a split.
<code>bootstrap</code>	True	Whether bootstrap samples are used.

Appendix B: Galaxy Clusters Catalog

Table B.1: Properties of the 42 mock galaxy clusters selected for this work. The columns represent: Cluster name, Right Ascension (RA), Declination (DEC), redshift (z), virial mass (M_{200}), virial radius (r_{200}), and the total number of member galaxies within $5 r_{200}$.

Cluster Name	RA hh:mm:ss	DEC hh:mm:ss	z	M_{200} $10^{14} M_{\odot}$	r_{200} Mpc	N_{mem} ($< 5 r_{200}$)
Abell 85	23:15:36.72	-27:01:20.86	0.058	8.32	1.71	1269
Abell 119	11:51:48.72	-07:20:08.80	0.039	8.30	1.71	1290
Abell 133	07:03:18.05	-41:02:46.99	0.052	4.02	1.34	735
Abell 147	11:27:00.19	+12:27:00.02	0.039	1.26	0.91	152
Abell 151	05:58:43.00	+15:18:51.84	0.051	2.42	1.13	379
Abell 168	02:50:34.69	-43:34:10.50	0.048	2.36	1.12	271
Abell 496	23:14:38.72	-19:32:46.72	0.031	5.64	1.49	1680
Abell 548	14:07:58.21	-35:03:02.73	0.034	2.90	1.19	361
Abell 754	19:47:23.00	-07:38:54.16	0.057	16.24	2.14	2421
Abell 780	16:41:53.56	+17:21:02.19	0.060	6.80	1.60	1278
Abell 957	00:38:23.73	+20:14:13.31	0.042	2.34	1.12	281
Abell 970	03:00:41.29	+45:40:34.63	0.059	4.02	1.34	612
Abell 1069	03:17:57.07	+44:28:35.13	0.062	1.41	0.95	221
Abell 1631	12:18:42.99	+03:40:10.07	0.045	2.54	1.15	273
Abell 1644	09:10:24.37	-68:22:29.01	0.049	6.65	1.59	692
Abell 2399	12:25:41.53	-00:54:09.95	0.049	3.33	1.26	718
Abell 2415	21:57:54.53	-42:21:24.12	0.057	1.40	0.94	264
Abell 2457	06:09:37.91	+20:26:12.34	0.057	2.00	1.06	240
Abell 2717	04:12:40.92	+34:03:33.63	0.046	1.49	0.96	166
Abell 2734	10:13:32.39	-77:59:42.94	0.059	2.42	1.13	215
Abell 2870	15:18:18.16	-40:45:19.60	0.015	4.69	1.40	693
Abell 2877	08:36:17.97	+23:56:22.77	0.018	1.31	0.91	204
Abell 3223	09:37:46.88	+28:58:03.90	0.055	3.10	1.23	354
Abell 3301	06:20:15.68	-73:35:59.31	0.045	2.34	1.12	202
Abell 3341	11:10:37.94	+61:17:07.33	0.040	1.98	1.06	244
Abell 3376	10:43:04.14	-40:34:05.06	0.036	4.16	1.36	577
Abell 3391	01:14:07.76	-70:43:27.17	0.051	4.64	1.41	425
Abell 3395	22:39:11.01	-39:25:35.61	0.054	4.77	1.42	501
Abell 3490	23:32:35.51	-50:14:44.21	0.065	3.16	1.24	415
Abell 3497	04:01:21.14	-25:45:31.26	0.059	2.87	1.20	377
Abell 3526	16:38:15.70	-34:38:54.05	0.014	3.18	1.23	511
Abell 3565	03:32:22.35	-35:00:25.41	0.016	2.01	1.06	268
Abell 3574	21:48:20.20	-07:43:27.27	0.019	2.97	1.20	856
Abell 3651	11:00:53.31	-34:59:28.52	0.058	3.05	1.22	400
Abell 3667	16:53:23.11	+39:54:14.32	0.047	8.76	1.74	1143
Abell 3716	09:10:16.22	-10:12:35.23	0.049	4.59	1.40	661
Abell 4038	06:49:21.40	+73:46:58.38	0.020	3.74	1.30	281
Abell 4059	11:33:33.11	+12:46:09.13	0.039	6.23	1.55	947
Antlia	04:14:02.57	+43:27:32.84	0.014	1.19	0.89	152
IIZw108	00:56:08.94	+02:28:41.20	0.046	1.79	1.02	273
MKW4	08:15:05.61	+18:54:16.59	0.011	1.54	0.96	194
MKW8	10:13:29.92	-21:37:52.49	0.031	1.48	0.95	176

Appendix C: Phase Space and 3D Distribution of Mock Clusters

This appendix presents the visual characterization of the 42 selected mock clusters. For each cluster, two panels are shown:

- **Left Panel:** The Project Phase Space diagram, showing the peculiar velocity v_{pec} versus projected radius r_{proj} .
- **Right Panel:** The 3D spatial distribution of galaxies (x, y, z) in the simulation.

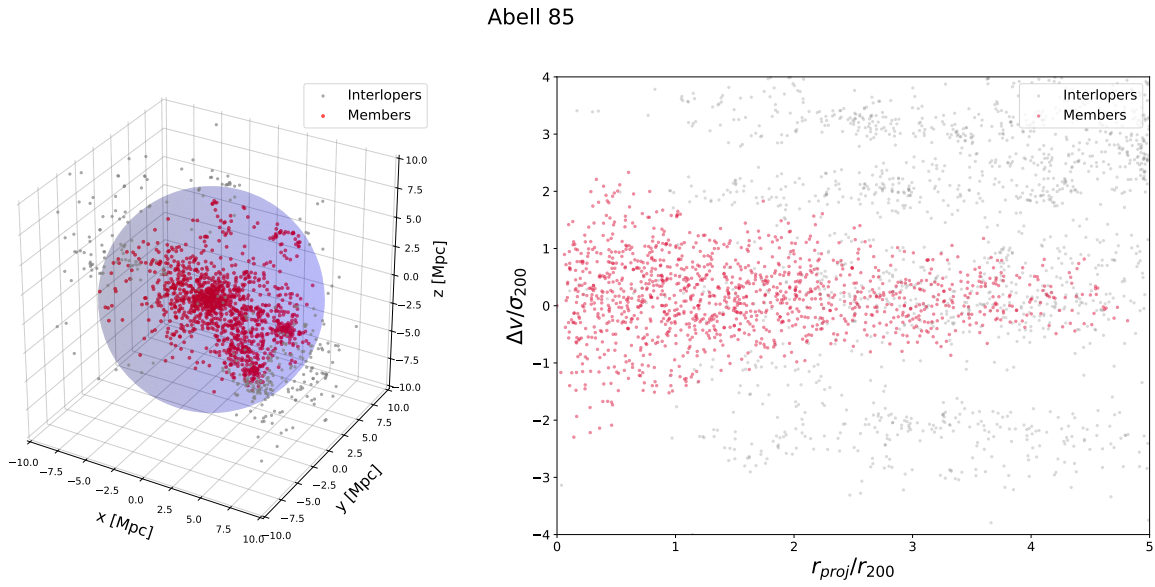


Figure C.1: Visual characterization of **Abell 85**

Abell 119

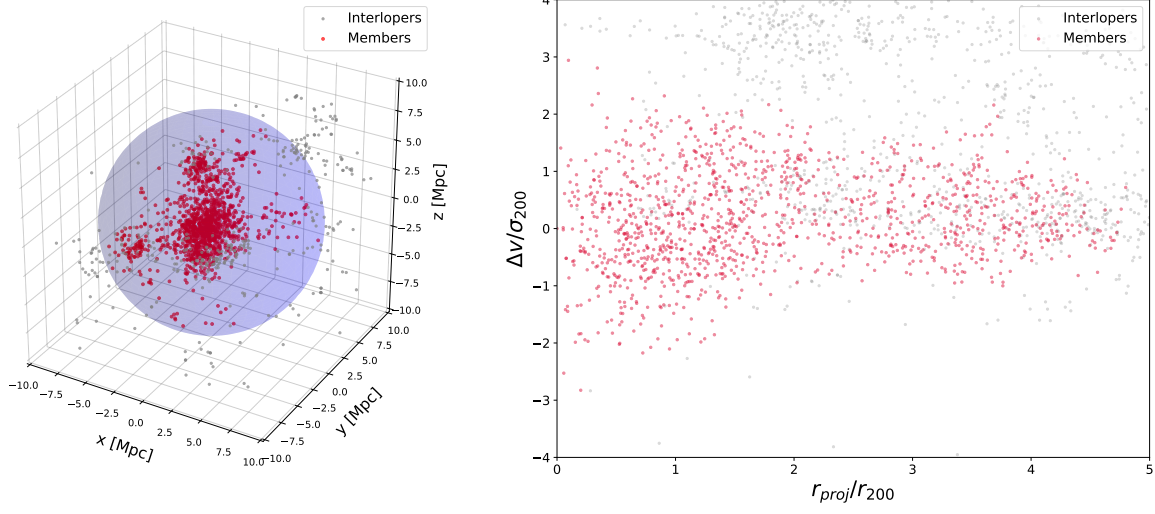


Figure C.2: Visual characterization of **Abell 119**

Abell 133

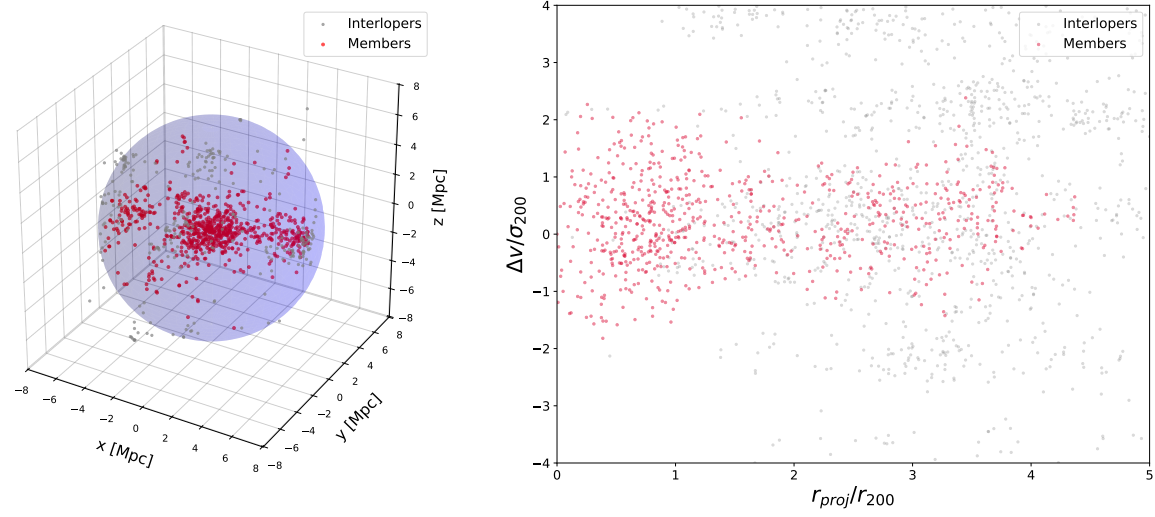


Figure C.3: Visual characterization of **Abell 133**

Abell 147

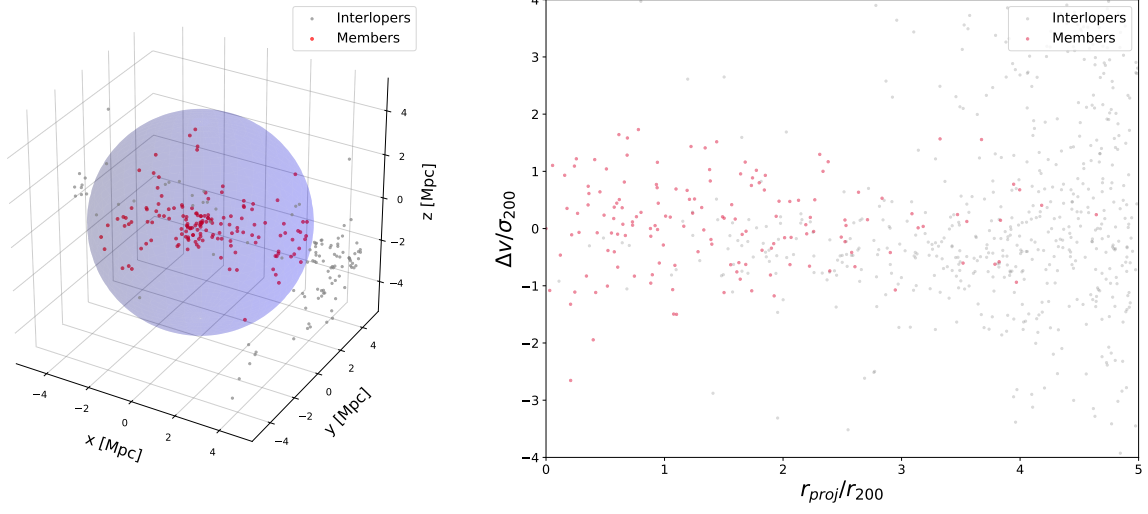


Figure C.4: Visual characterization of **Abell 147**

Abell 151

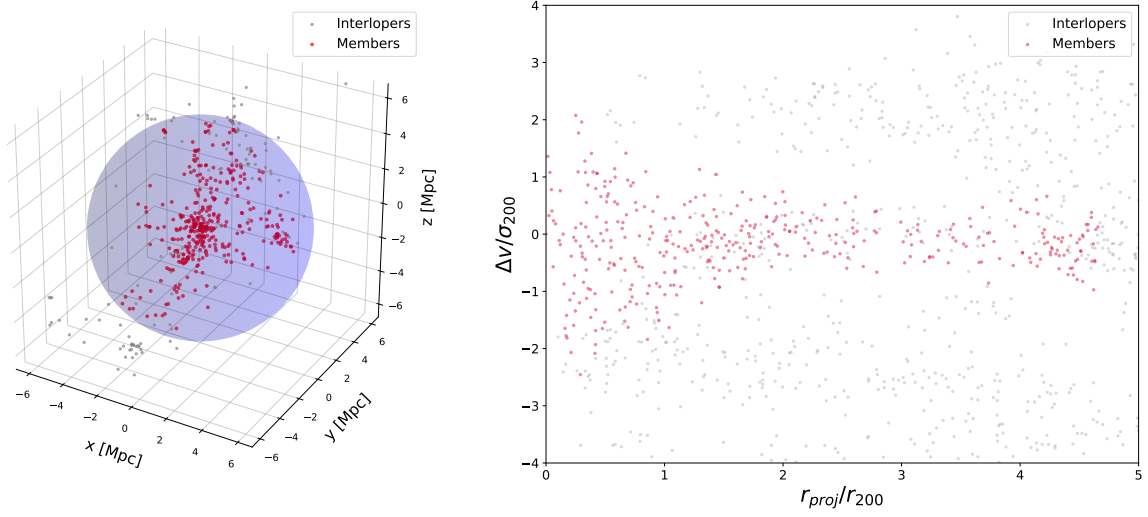


Figure C.5: Visual characterization of **Abell 151**

Abell 168

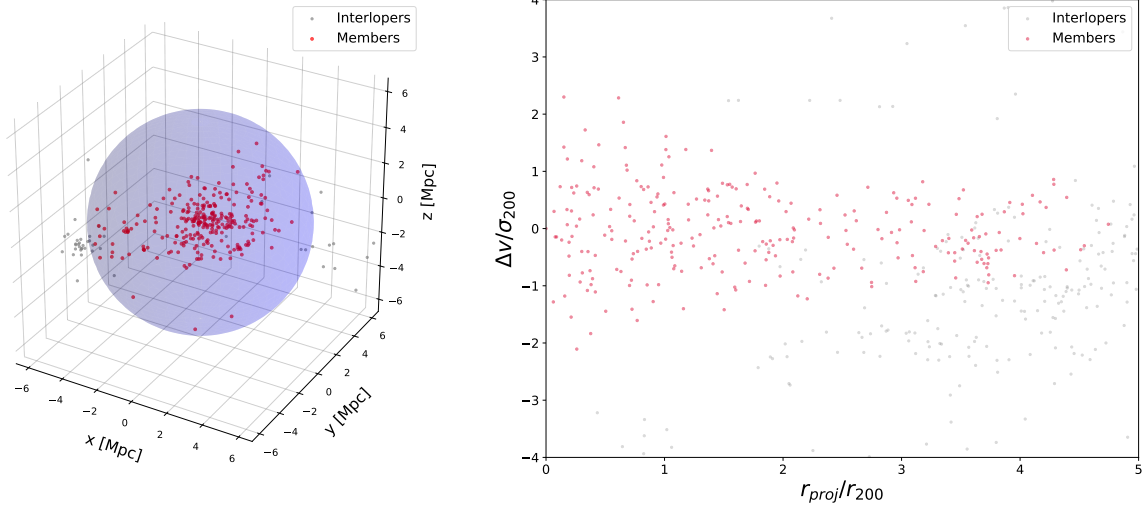


Figure C.6: Visual characterization of **Abell 168**

Abell 496

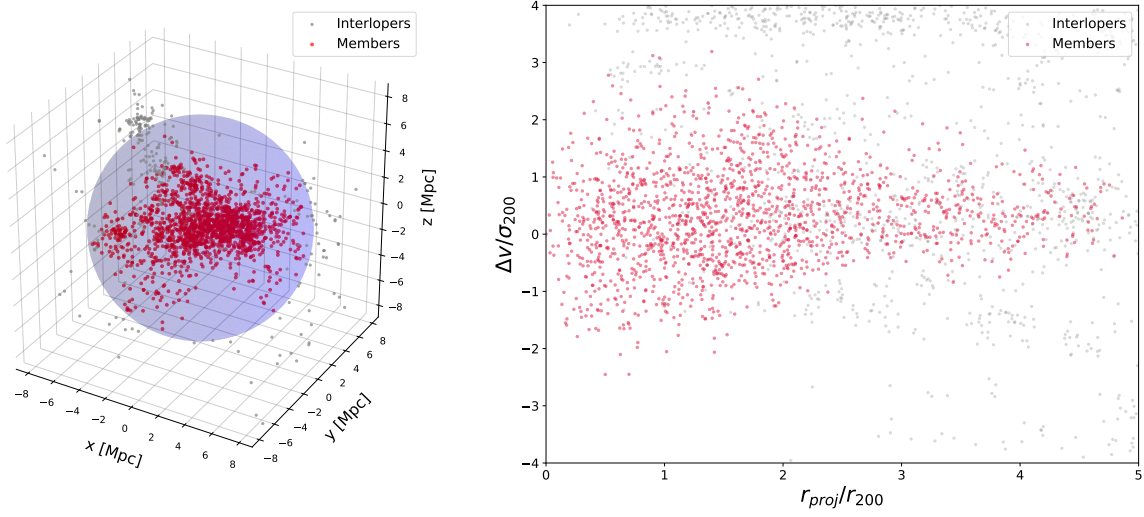


Figure C.7: Visual characterization of **Abell 496**

Abell 548

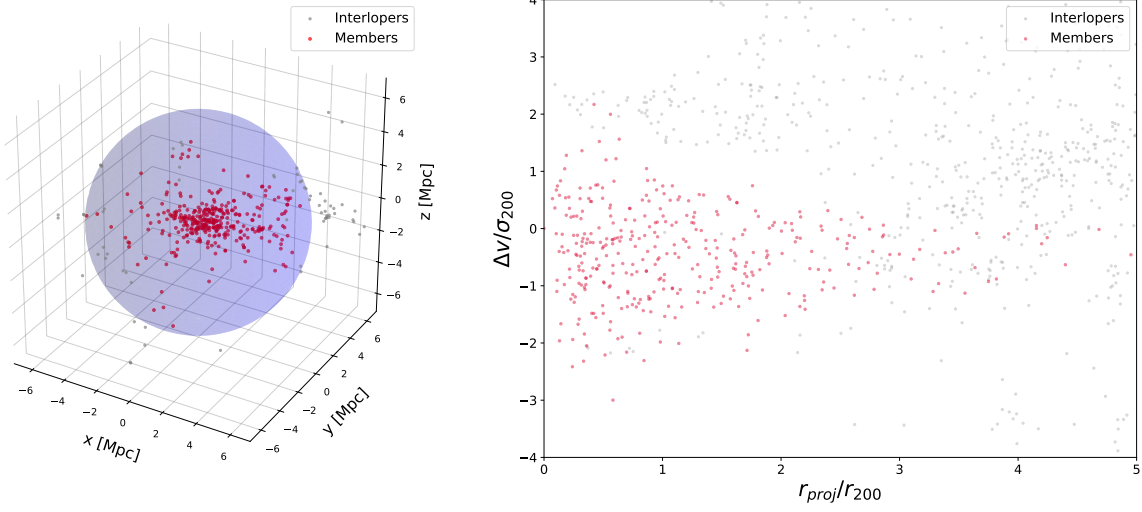


Figure C.8: Visual characterization of **Abell 548**

Abell 754

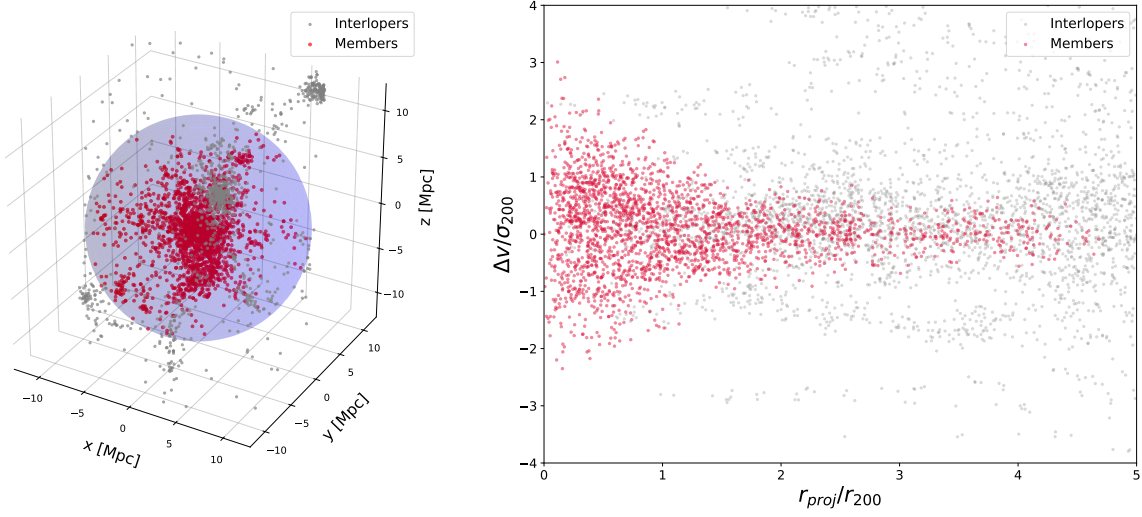


Figure C.9: Visual characterization of **Abell 754**

Abell 780

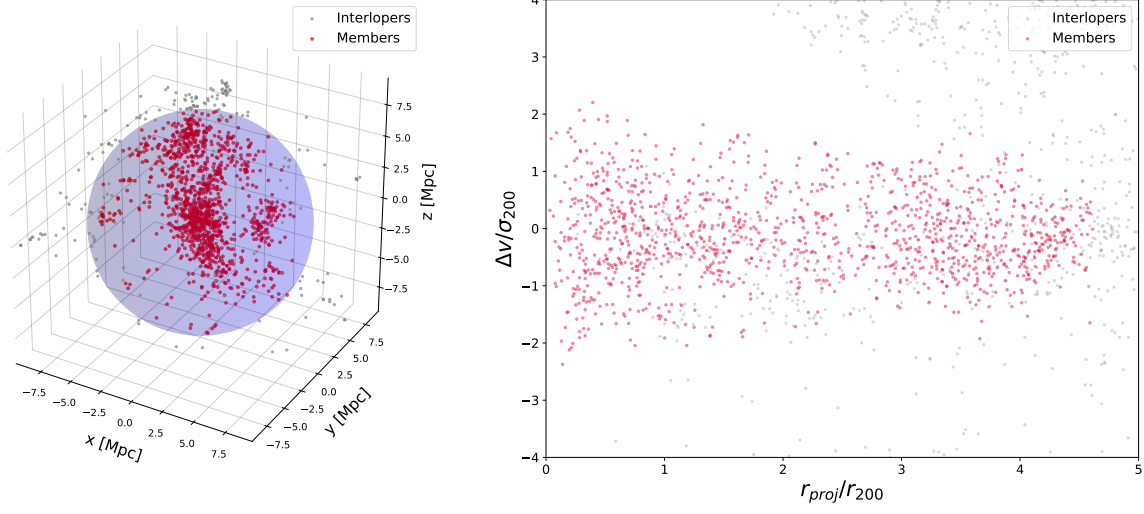


Figure C.10: Visual characterization of **Abell 780**

Abell 957

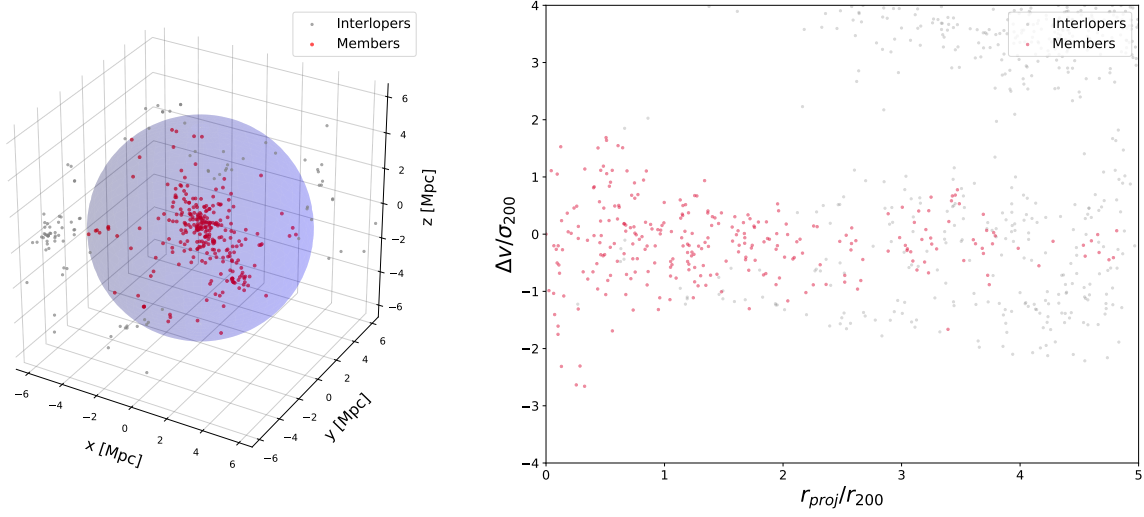


Figure C.11: Visual characterization of **Abell 957**

Abell 970

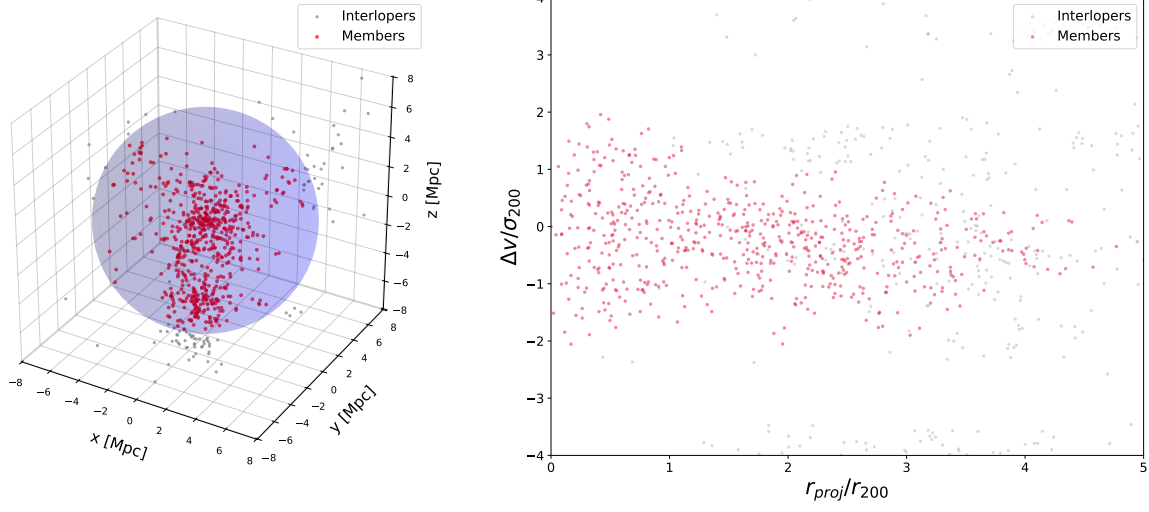


Figure C.12: Visual characterization of **Abell 970**

Abell 1069

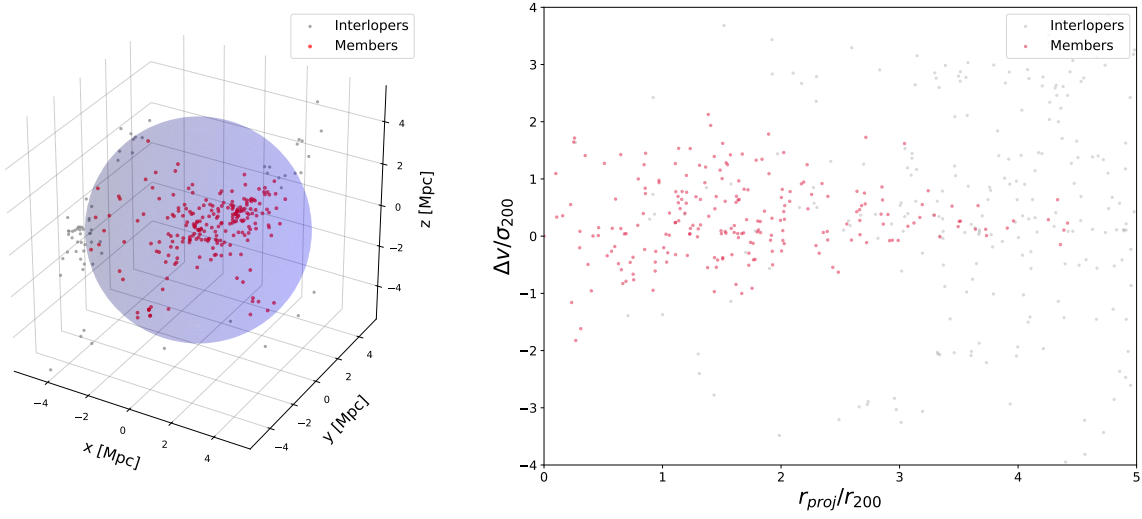


Figure C.13: Visual characterization of **Abell 1069**

Abell 1631

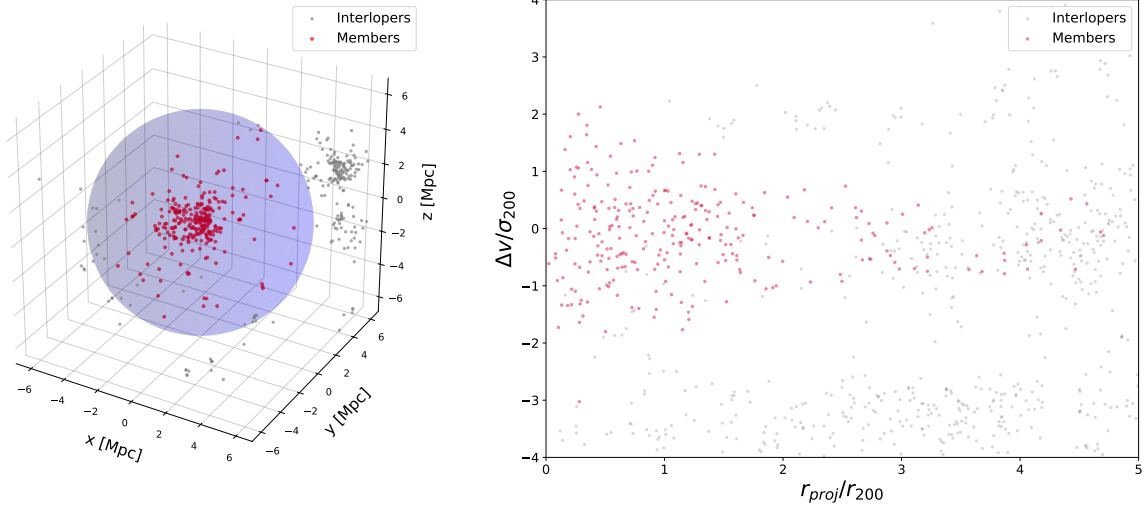


Figure C.14: Visual characterization of **Abell 1631**

Abell 1644

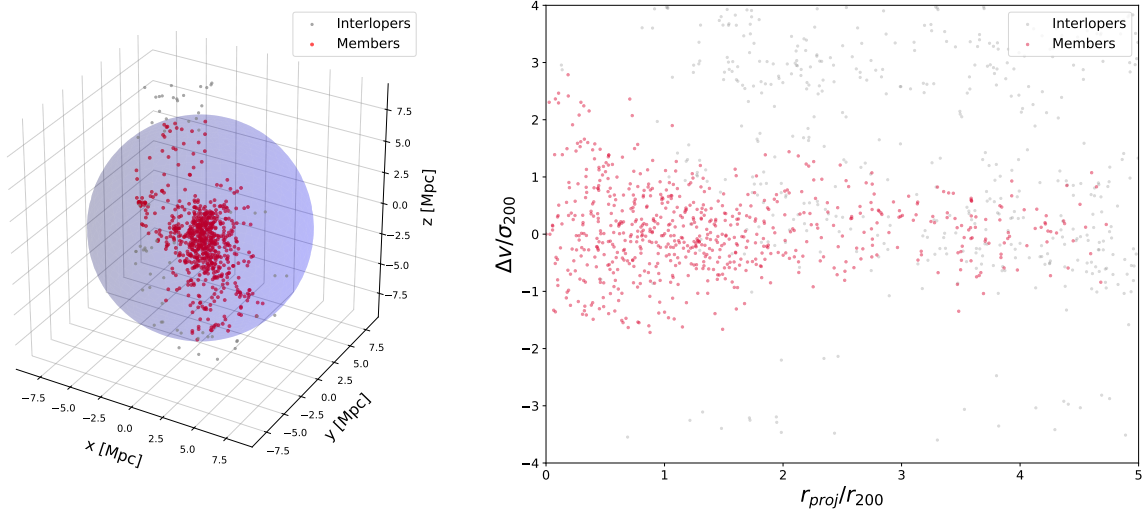


Figure C.15: Visual characterization of **Abell 1644**

Abell 2399

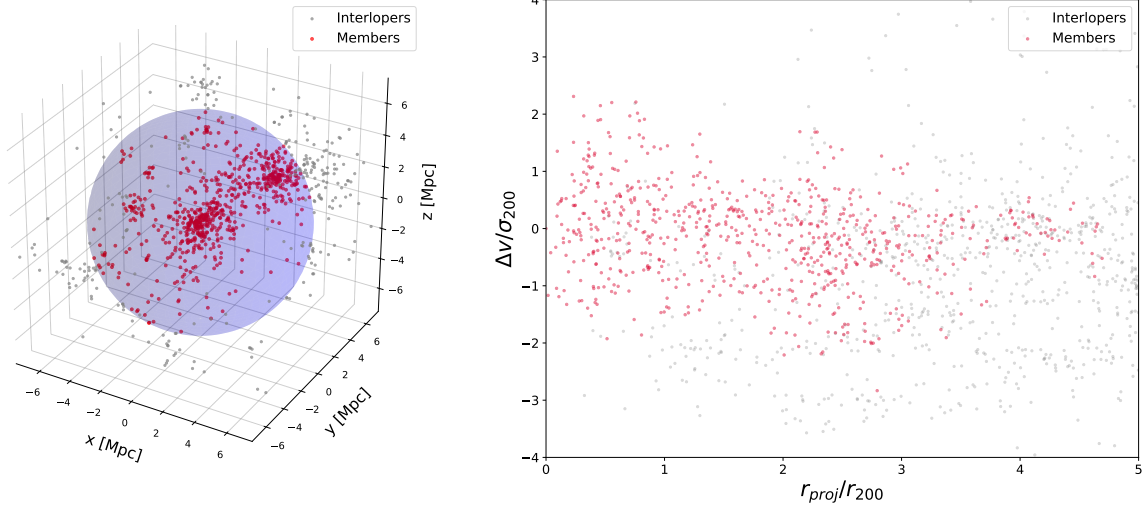


Figure C.16: Visual characterization of **Abell 2399**

Abell 2415

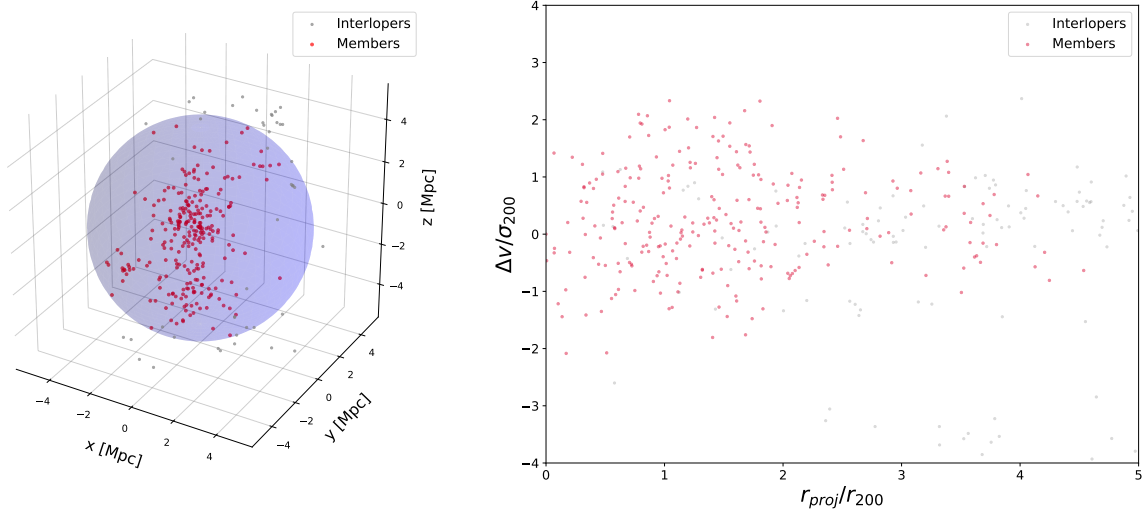


Figure C.17: Visual characterization of **Abell 2415**

Abell 2457

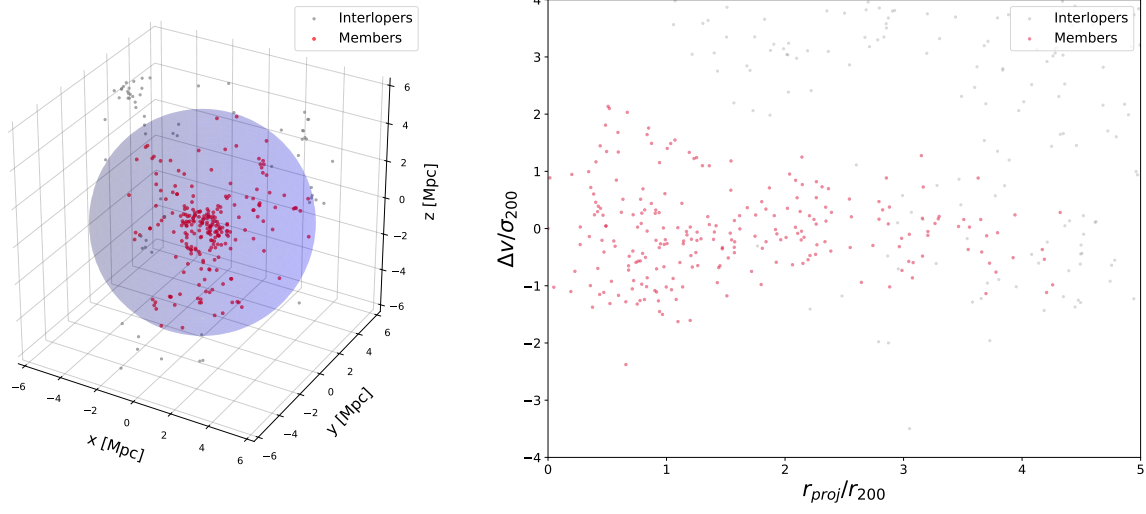


Figure C.18: Visual characterization of **Abell 2457**

Abell 2717

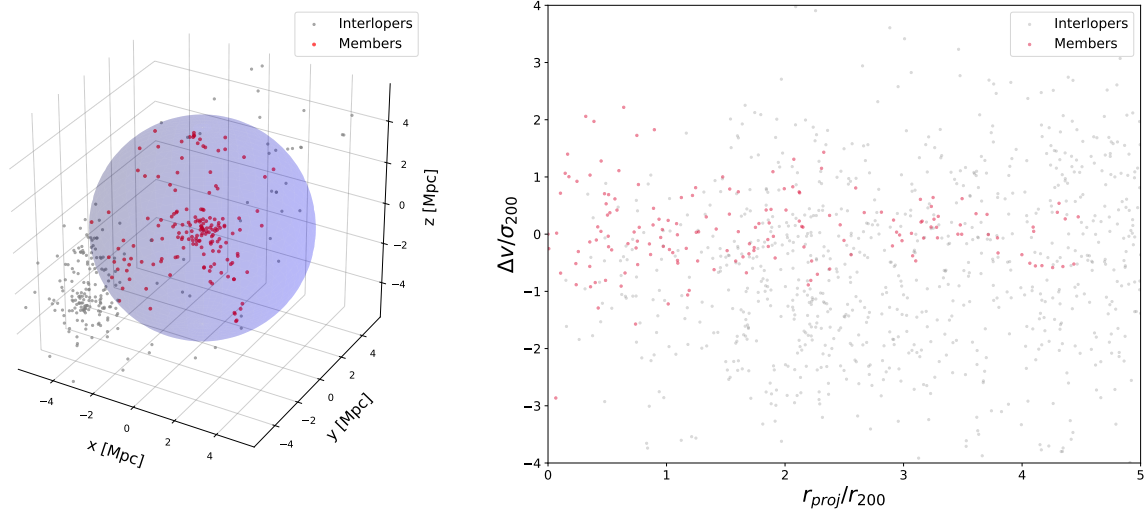


Figure C.19: Visual characterization of **Abell 2717**

Abell 2734

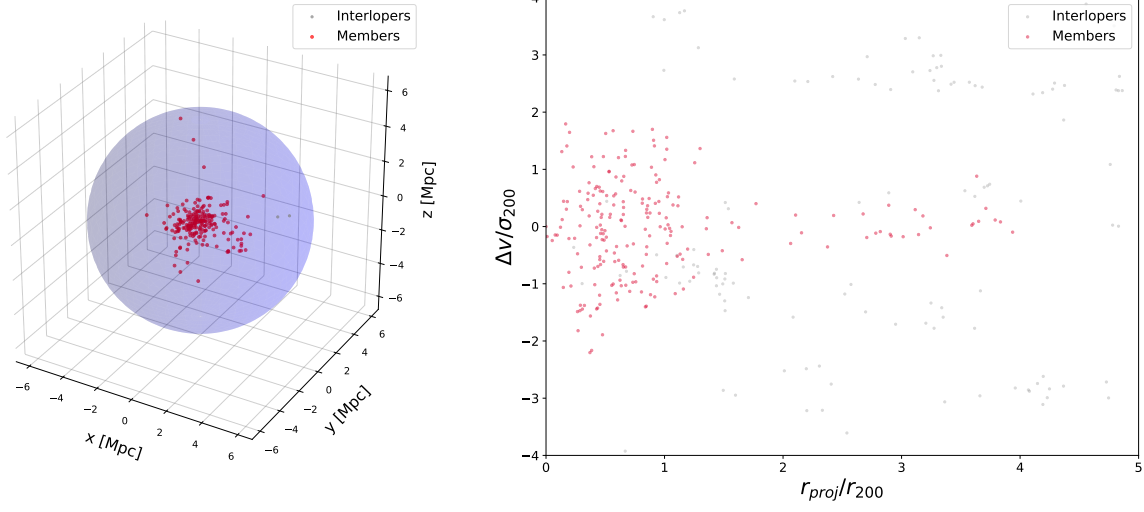


Figure C.20: Visual characterization of **Abell 2734**

Abell 2870

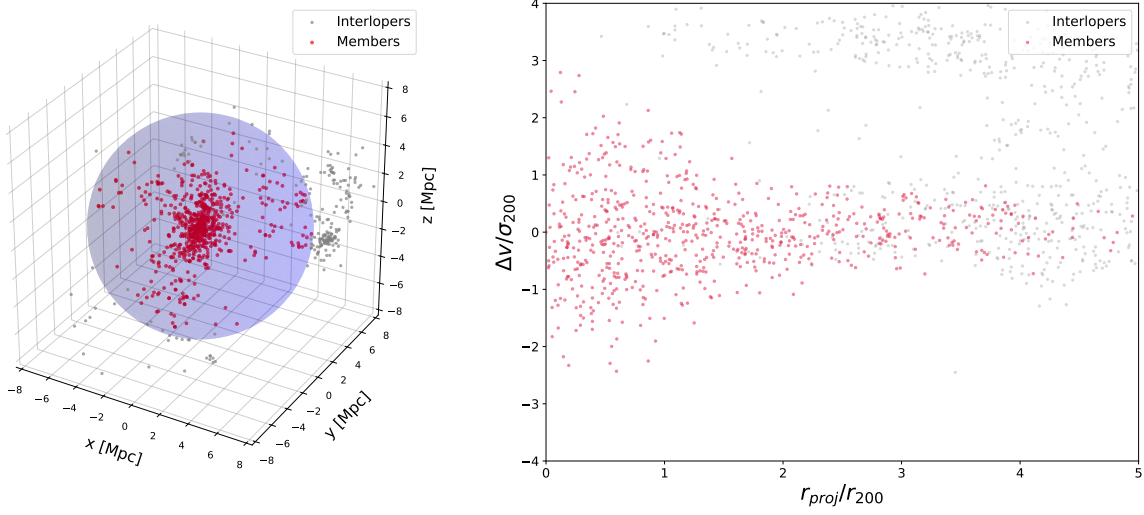


Figure C.21: Visual characterization of **Abell 2870**

Abell 2877

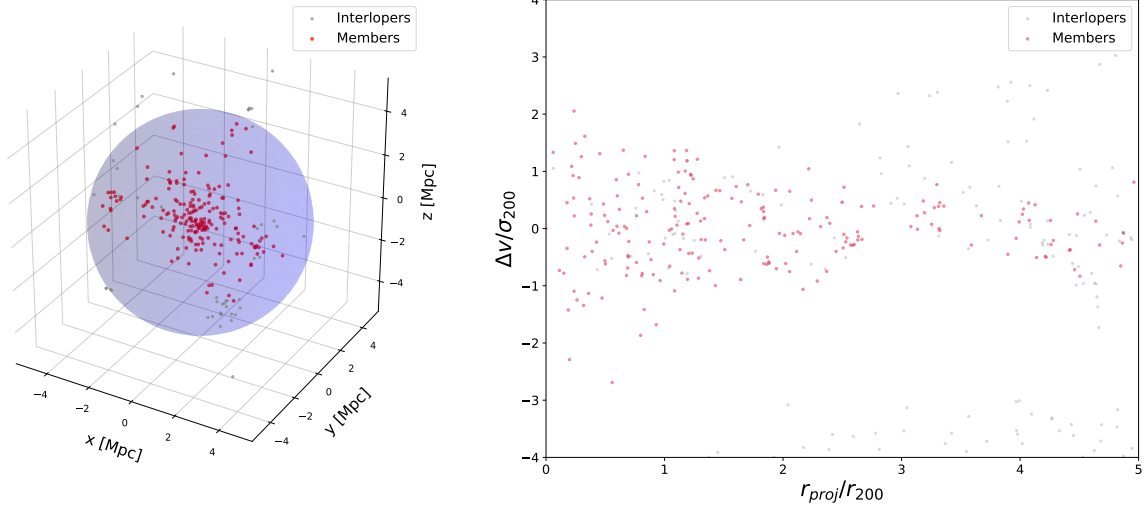


Figure C.22: Visual characterization of **Abell 2877**

Abell 3223

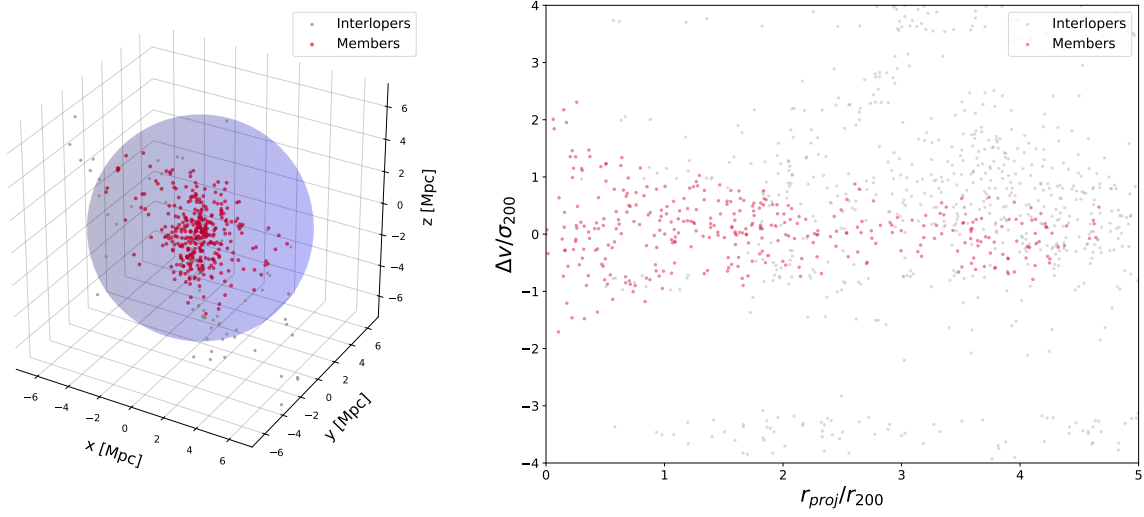


Figure C.23: Visual characterization of **Abell 3223**

Abell 3301

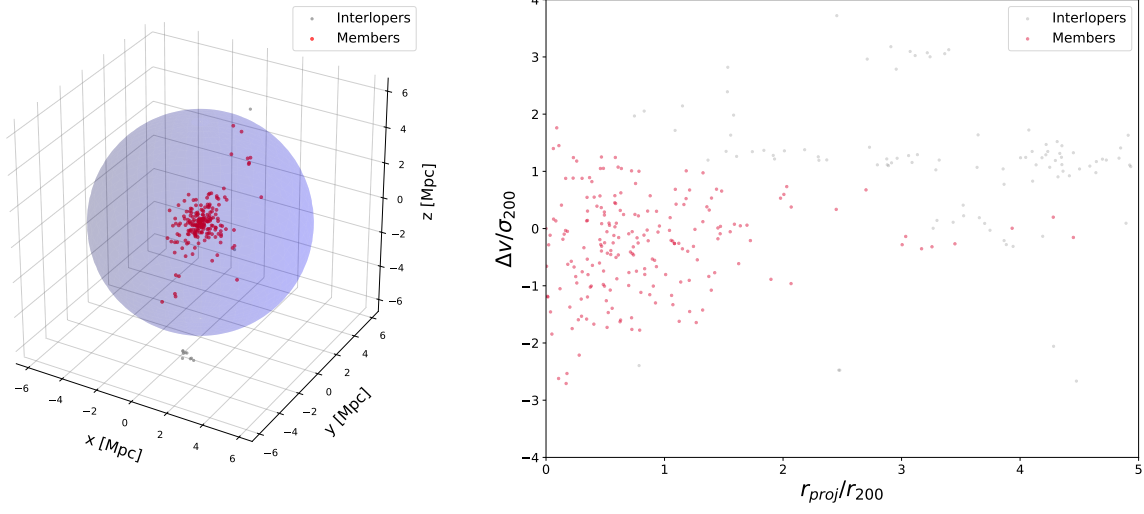


Figure C.24: Visual characterization of **Abell 3301**

Abell 3341

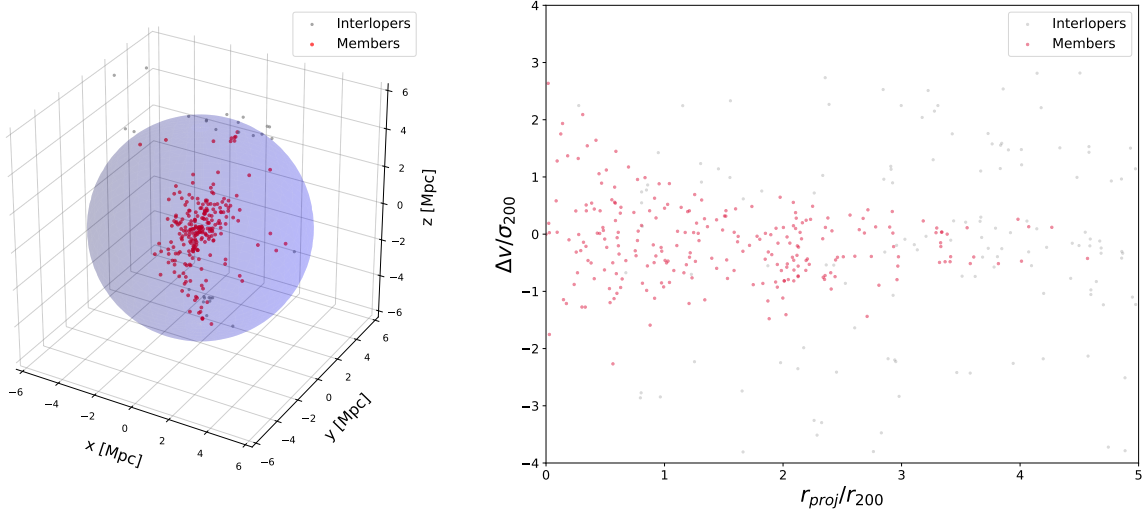


Figure C.25: Visual characterization of **Abell 3341**

Abell 3376

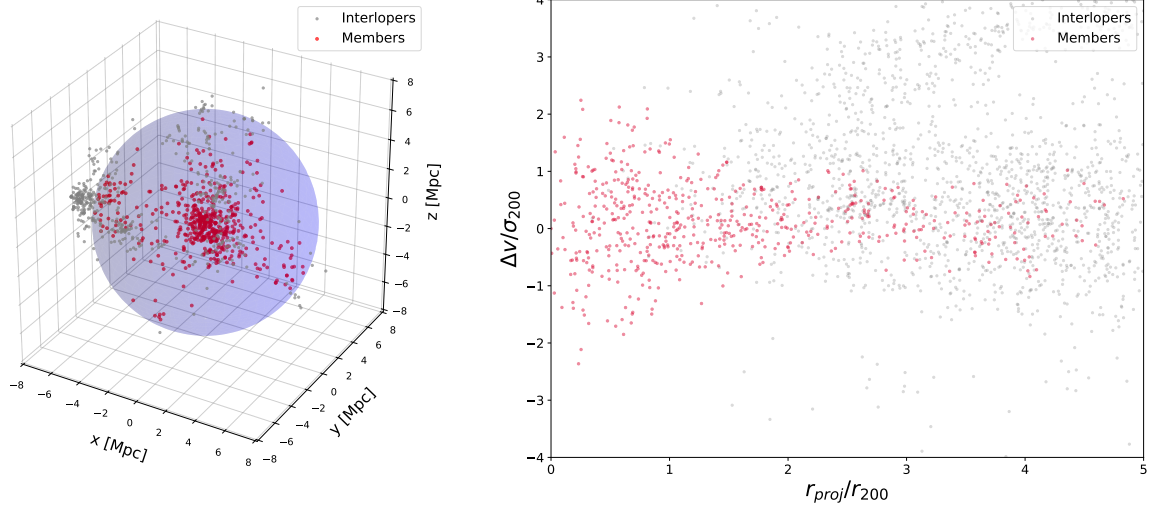


Figure C.26: Visual characterization of **Abell 3376**

Abell 3391

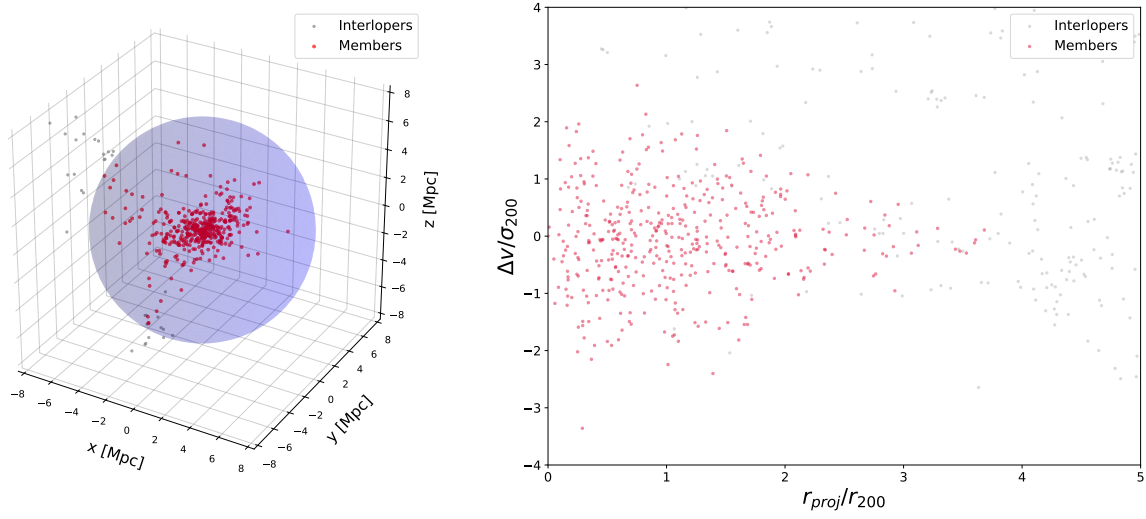


Figure C.27: Visual characterization of **Abell 3391**

Abell 3395

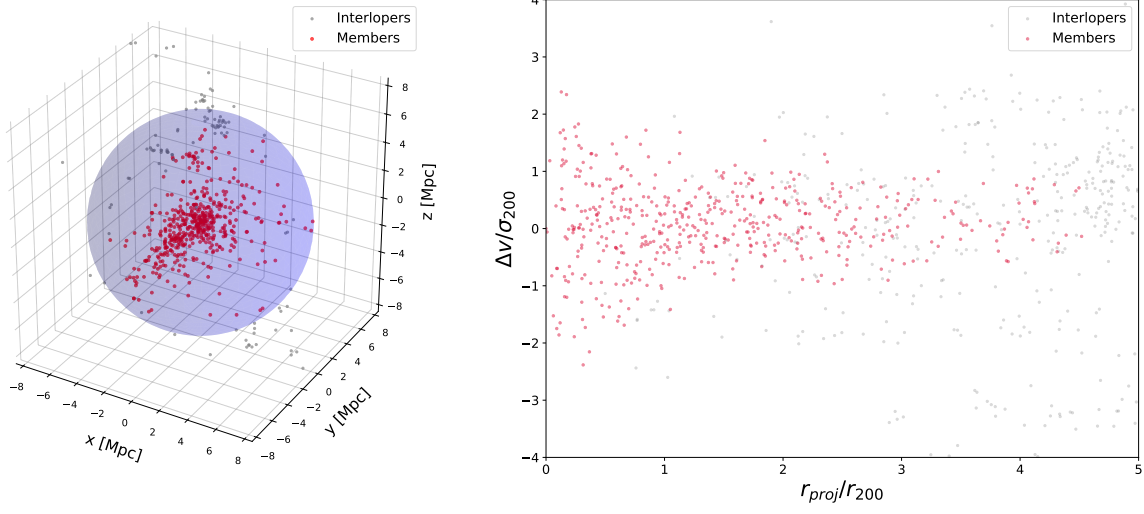


Figure C.28: Visual characterization of **Abell 3395**

Abell 3490

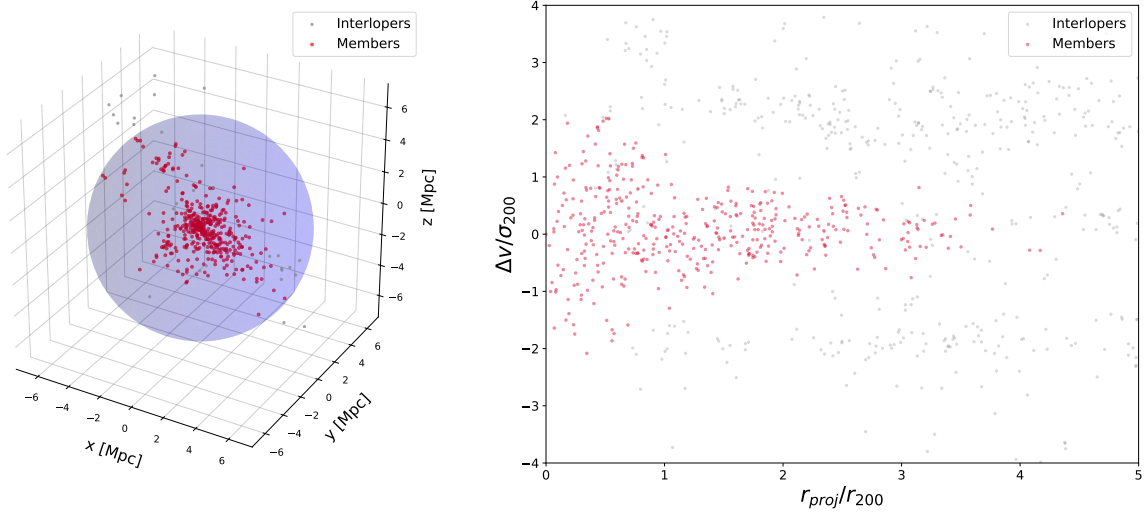


Figure C.29: Visual characterization of **Abell 3490**

Abell 3497

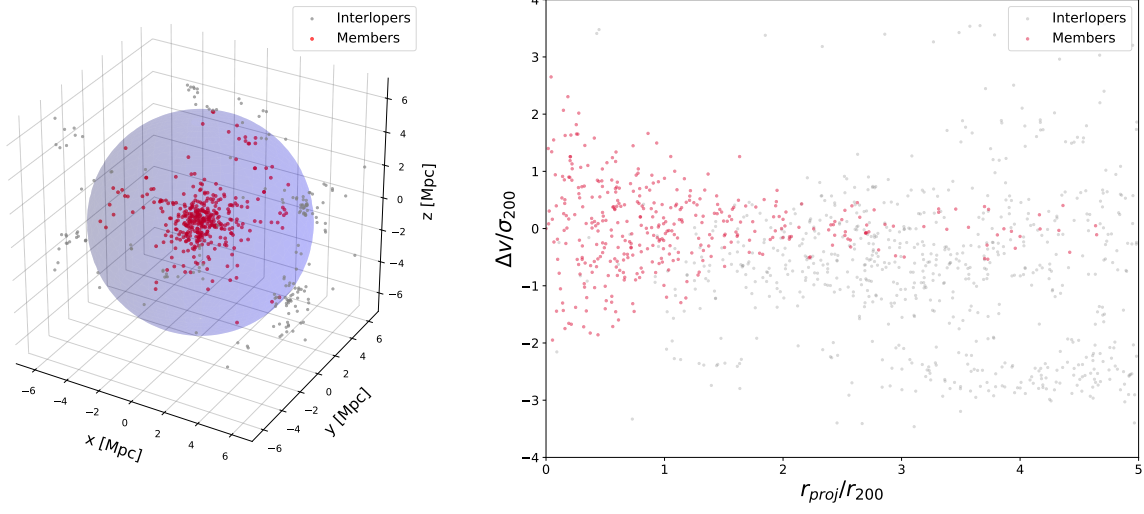


Figure C.30: Visual characterization of **Abell 3497**

Abell 3526

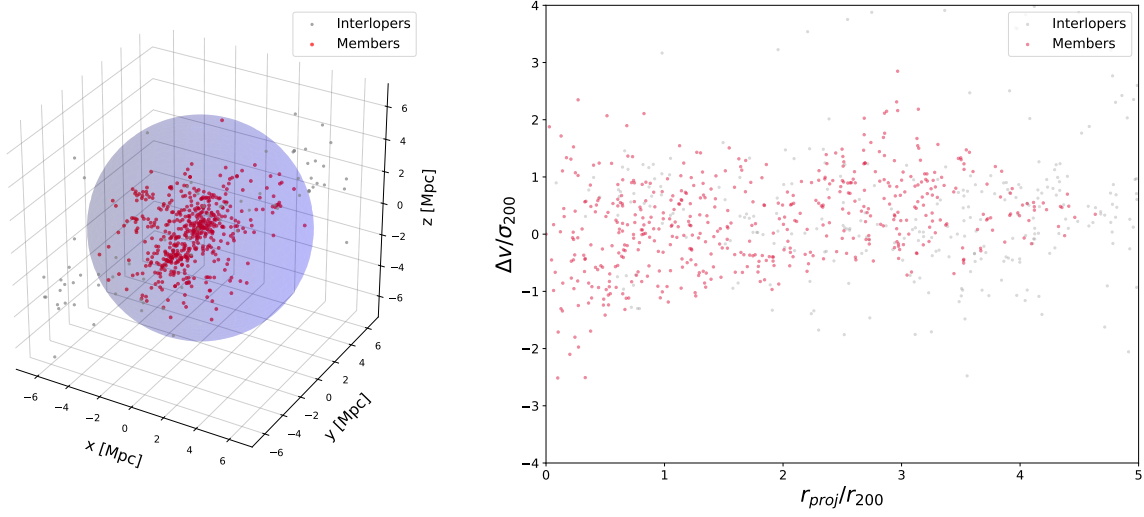


Figure C.31: Visual characterization of **Abell 3526**

Abell 3565

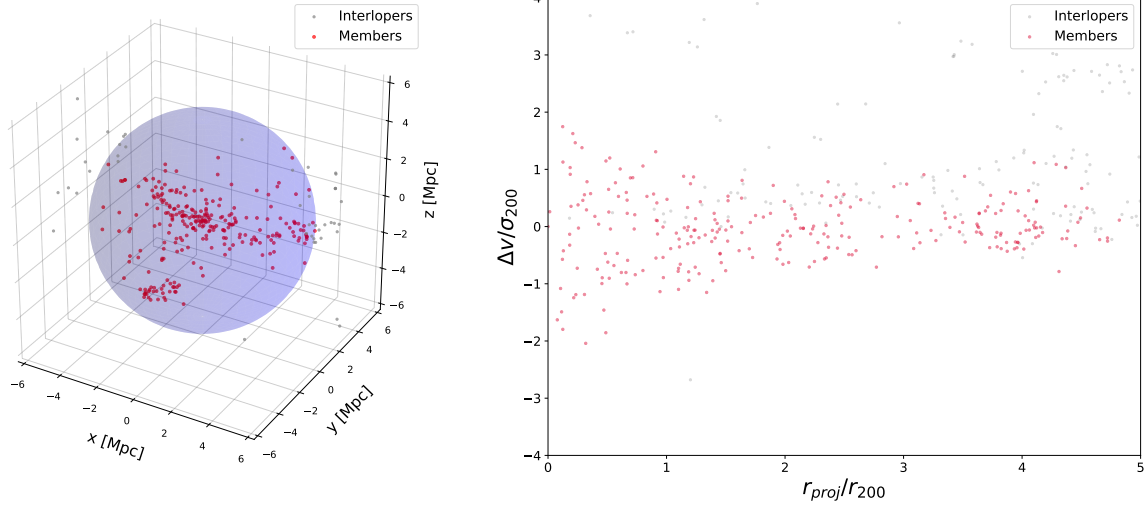


Figure C.32: Visual characterization of **Abell 3565**

Abell 3574

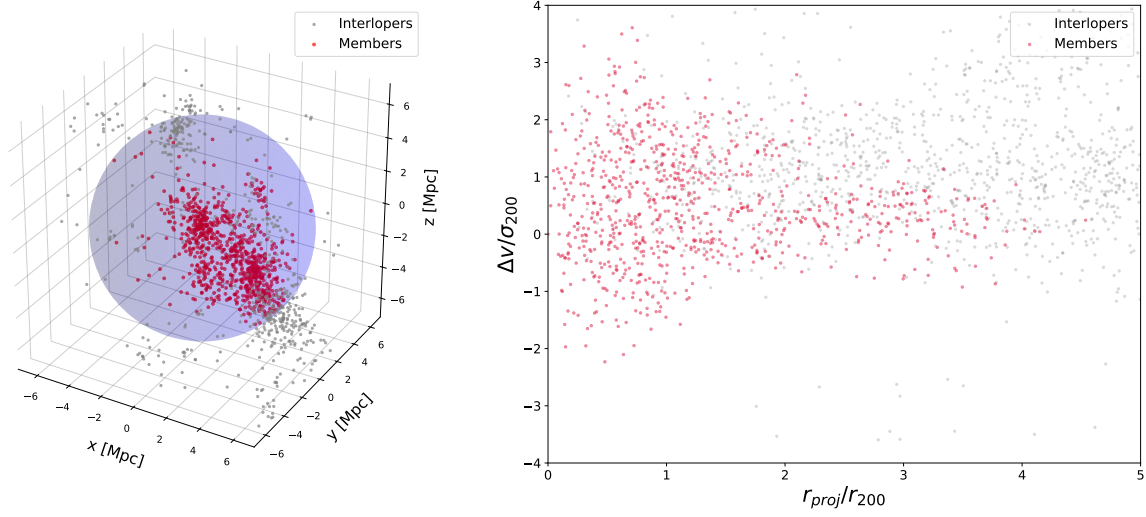


Figure C.33: Visual characterization of **Abell 3574**

Abell 3651

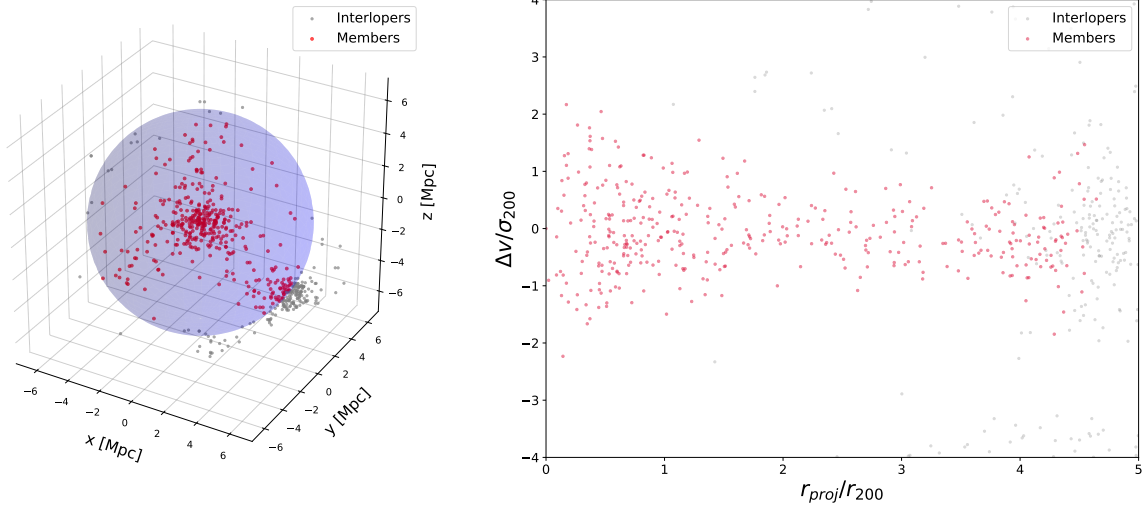


Figure C.34: Visual characterization of **Abell 3651**

Abell 3667

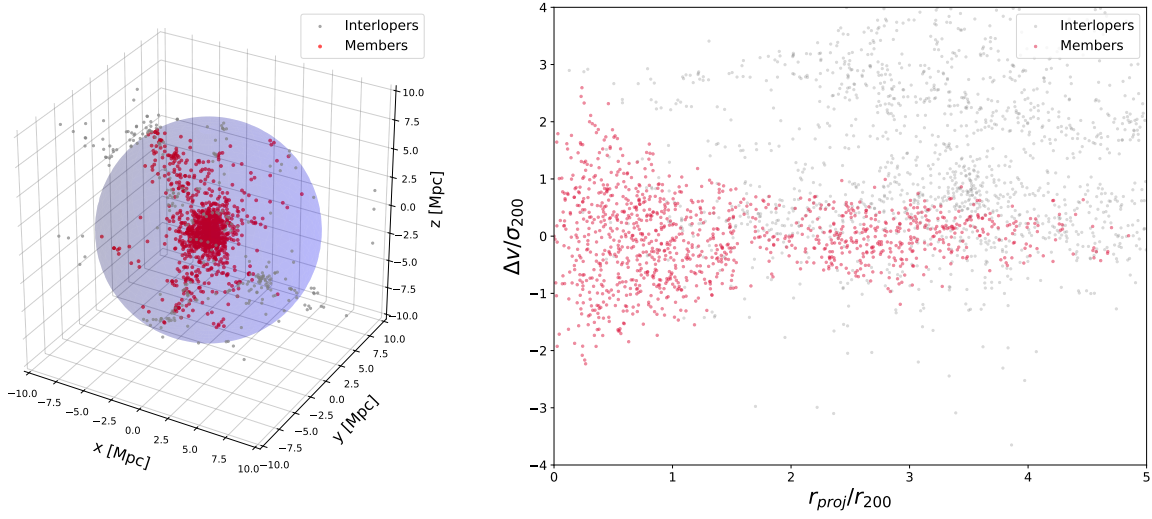


Figure C.35: Visual characterization of **Abell 3667**

Abell 3716

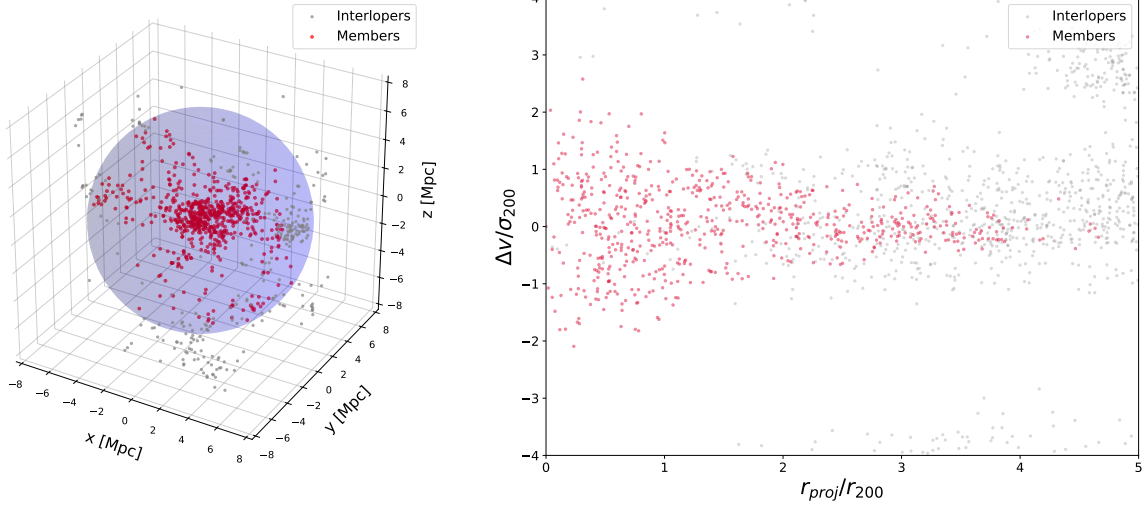


Figure C.36: Visual characterization of **Abell 3716**

Abell 4038

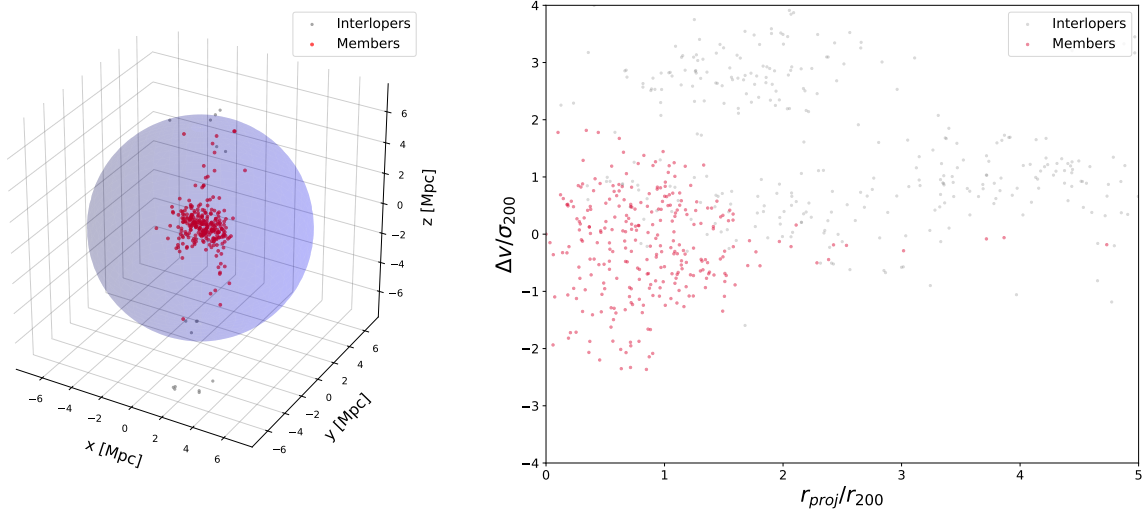


Figure C.37: Visual characterization of **Abell 4038**

Abell 4059

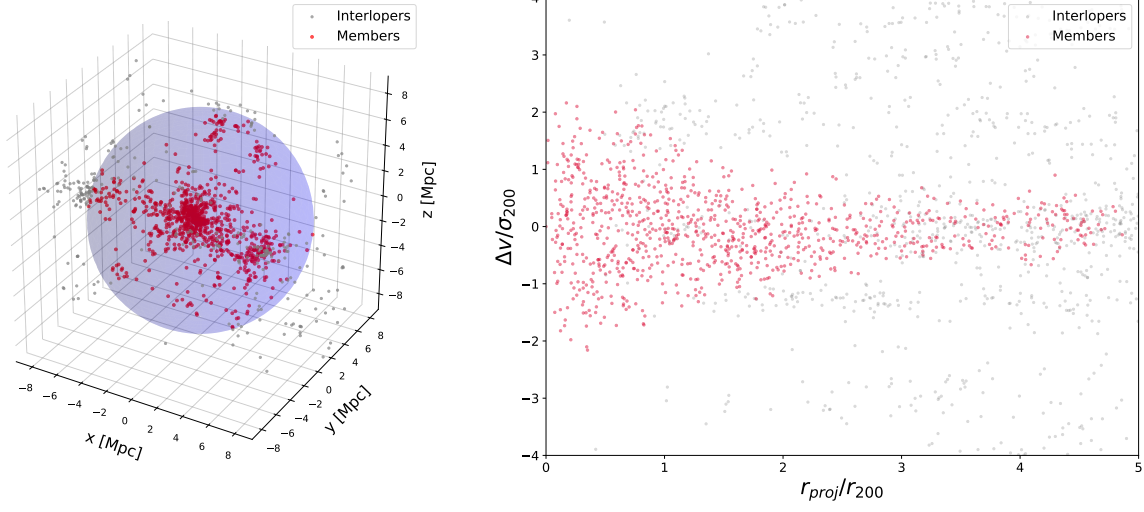


Figure C.38: Visual characterization of **Abell 4059**

Antlia

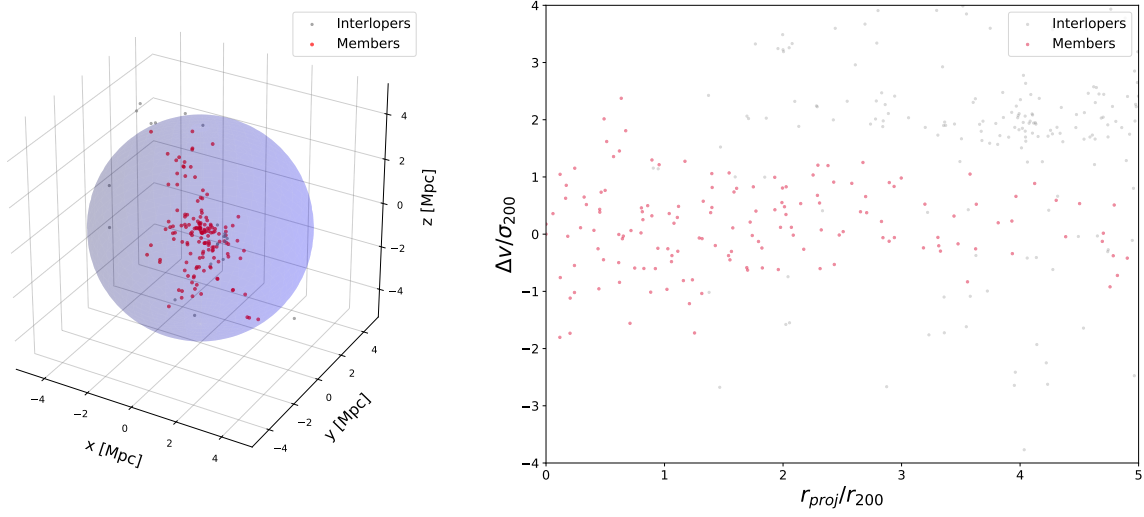


Figure C.39: Visual characterization of **Antlia**

IIZw108

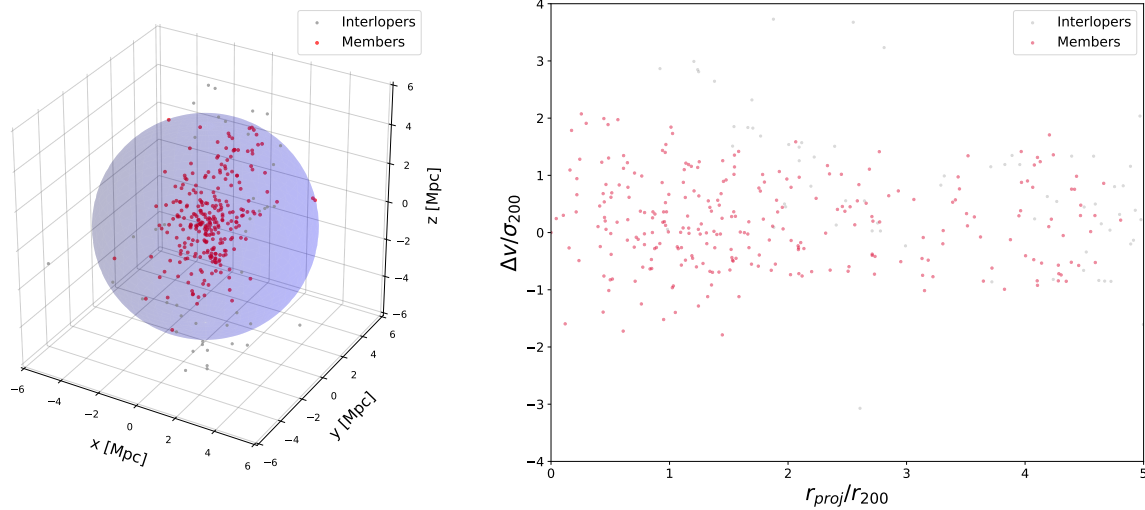


Figure C.40: Visual characterization of **IIZw108**

MKW4

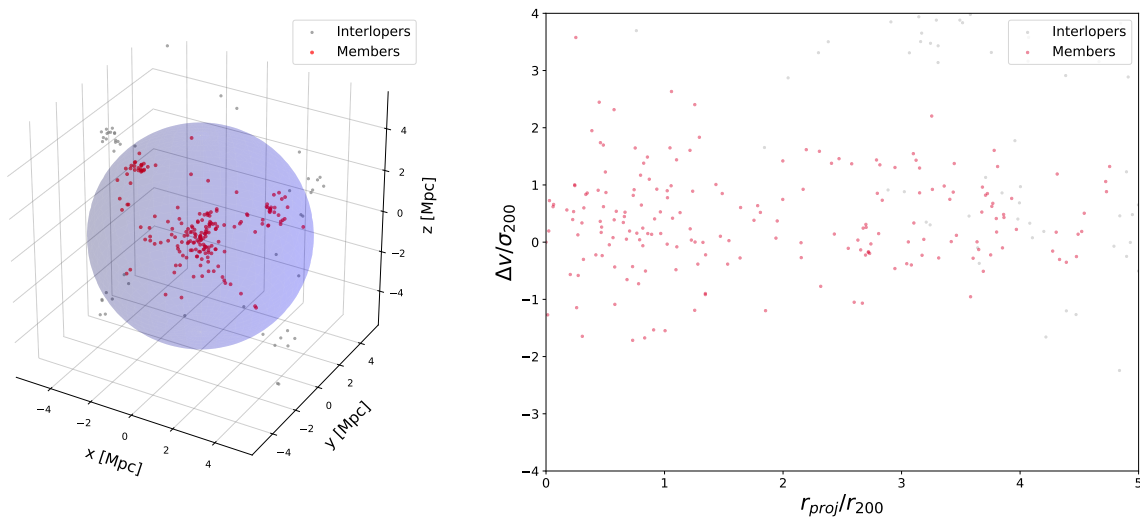


Figure C.41: Visual characterization of **MKW4**

MKW8

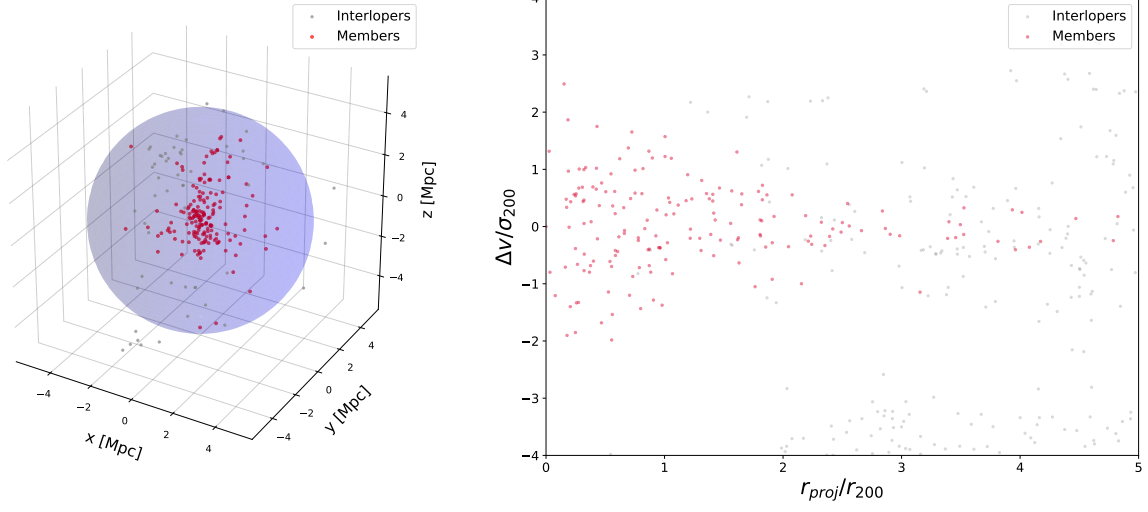


Figure C.42: Visual characterization of **MKW8**